Engineering Notebook



4613R

Team Number

Barker Redbacks

Team Name

Barker College

School

11/06/2026

Start Date

VEX ROBOTICS

REC FOUNDATION

Date 23/8/2026 Event Name Vex V5 Shore Regionals

Innovate Award Submission Information Form

Instructions for team: Please fill out all information, printing clearly. This form should be included immediately after the Engineering Notebook's cover page. In the case of physical notebooks, this form can be printed out and placed in the notebook. For digital notebooks, this form can be scanned in and included. Teams may only submit **one** aspect of their design to be considered for this award at each event. Submission of multiple aspects will nullify the team's consideration for this award.

Full Team Number: 4613R

Brief description of the novel aspect of the team's design:

We have incorporated a chain bar on our robot, which allows us to score from both the front and back of the robot, improving our offensive ability and providing greater flexibility during matches. This allows us to approach scoring areas from different directions without needing to reposition the entire robot, saving valuable time and making our movements more efficient. Additionally, the chain bar can hold both cups and pins, allowing us to collect, transport and prepare multiple game objects for scoring. When combined with our elevator, this enables the robot to move game objects efficiently to the required height, allowing for faster and more consistent scoring throughout the match.

Identify the page numbers and/or the section(s) where documentation of the development of this aspect can be found:

Two-Bar Arm - Operation and Evidence - page 65

Explain why your submission is unique from other approaches to the problem it solves or task it performs:

This is important as the game is played in 'cycles' where we move from the match loader or holding other pins and score them, and the game is just a repetition of that cycle. Our front to back scoring method means that we do need to spend a large amount of time turning around, using this innovative strategy we have decreased our individual 'cycle' time to score more. However, other teams may not have included this design due to the misguided opinion that turning the robot around is much easier than bringing the pin over the robot itself.

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ENGINEERING NOTEBOOK STRUCTURE / FORMAT

How the work flows.

Structure

----------Format----------

Goal

To establish a clear and consistent format that effectively documents our design process, decisions, testing and progress throughout the season.

Identify
• Finding challenges in the game and robot design. • States the goals and solutions for the season.
Brainstorm
• Lists possible robot designs for the season and states the advantages and disadvantages of each robot design.
Select
• Explains reasoning for chosen robot design. • Highlights the robot design chosen.
CAD
• Displays different prototypes of mechanisms. • The design of custom parts.
Testing
• Records the steps of which we tested different mechanisms of the robot. • Displays solutions found during the testing process.
Programming
• Displays the programming journey. • Displays and explains vision scoring assists. • Explains the custom macros for scoring and intaking.

Format

----------Reasoning----------

Goal

To establish a clear and accessible Engineering Notebook structure that accurately documents our design process, collaboration, decisions, testing and progress throughout the season.

Formatting
• The notebook is formatted to strictly follow the judge's rubric and the engineering design process. • This allows us to stay organized and on track, clearly document our process and ensure no aspects of the design process are overlooked.
Organisation
• The notebook is organised for each section to have a dedicated tab at the beginning. • This allows clear separation between different stages and keeps sections in chronological order.
Sections
• Sections are aimed to follow the engineering design process, showing progression throughout the season. • Each section has a clear purpose, allowing clear understanding of design decisions, reasoning and progress.
Online Notebook
• Allows collaboration across the entire team and makes the notebook accessible so all members can contribute, edit and review information. • An online notebook also supports visual documentation, for example images and graphs can be inserted alongside analysis.
Maintenance
• Signing and dating each page helps maintain the notebook in chronological order. • It also shows when information, decisions and design developments were documented.

TEAM STRUCTURE

How we effectively work together to create something cohesive.

Team Collaboration

----------Structure----------

Goal

To create an inclusive and collaborative team environment where every member contributes equally, stays informed, has access to shared resources and feels valued throughout the robotics season.

Goals

- **Equal Contribution:** Ensure every team member feels valued and has meaningful responsibilities within the team.
- **Equal Workload:** Distribute tasks so that no single team member is responsible for most of the work.
- **Daily Communication:** Keep all members updated about the robot's progress, changes and current tasks.
- **Shared Access:** Ensure all digital resources including Engineering Notebook, CAD and code are accessible to every member.

Methods

- **Regular Feedback:** Collect feedback from team members to identify what is working well and what improvements can be made.
- **Online Communication:** Use an online communication platform for robotics discussions, updates and daily reports.
- **Shared Digital Platforms:** Use shared online platforms to organise and manage resources such as the Engineering Notebook, CAD files and programming code.
- **Task Allocation:** Clearly assign responsibilities to team members and rotate tasks to ensure that everyone gains experience across different areas of robotics.
- **Progress Updates:** Provide regular updates on design changes, testing results and upcoming tasks so the entire team remains informed and updated.

Team Communication

----------Updates----------

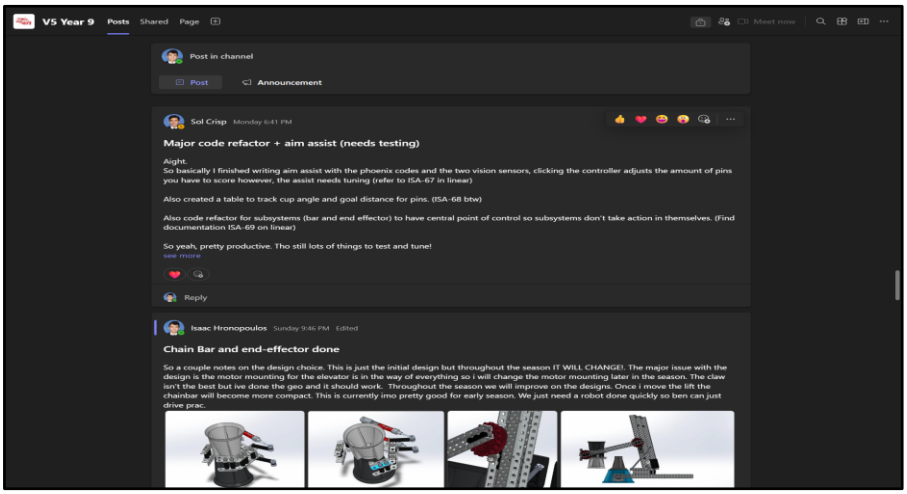
Goal

To establish an accessible and consistent communication system that keeps all team members informed, organised and aware of their responsibilities throughout the robotics season.

How It Is Achieved

- **Universal Accessibility:** We will use Microsoft Teams because it is accessible across different devices and operating systems, this allows all team members to communicate regardless of their hardware or software.
- **Daily Updates:** We will send daily messages to keep teammates informed and updated about our progress with the robot, including designing, building, testing and programming.
- **Daily To-Do Lists:** At the end of each day, we will use the chat to share clear to-do lists so every team member knows what needs to be completed and what they are responsible for.
- **Centralised Communication:** Teams will provide a central location for robotics-related communication, ensuring important updates and information can be easily accessed by everyone.

Example



DESIGN CYCLE 1

The first iteration.

IDENTIFY

The issue.

Season Goals

----------Identify----------

Goal

To establish clear robot, team and competition goals that guide our decisions and measure our process throughout the season.

Robot Goals

- Build a **simple, reliable and effective** robot.
- Consistently **control goal zones** during matches.
- Develop a **reliable mechanism** to intake game pieces from different angles and positions.
- Design the robot to **perform consistently** even against strong oppositions.
- Learn from other team's **strategy, designs and mechanisms** to improve our robot.

Team Goals

- Develop strong **communication and teamwork** throughout the season.
- Improve our understanding and knowledge of **robot design, strategy and engineering.**
- Learn from **competition experiences** especially at State and Nationals.
- Use competition experience to identify **areas for improvement.**
- Work effectively together as a team under **competition pressure.**

Competition Goals

- Remain competitive in areas including **robot design, strategy and documentation.**
- Continuously **improve the robot** throughout the competition season.
- **Learn** from successful teams especially at State and Nationals.
- **Adapt our strategy** based on what is observed from other teams.
- Achieve a high placement at Worlds through continuous **improvement and preparation.**
- Use every competition as an opportunity to **learn, test and improve.**

The Challenge

----------Identify----------

Goal

To analyse the Override game and identify the requirements, challenges and strategies needed to develop an effective and competitive robot

Understanding The Challenge

- Override (2026 – 2027) requires robots to complete multiple tasks **efficiently, consistently and strategically.**
- Teams must **analyse the game manual and field** before designing a robot.
- **Successful robots must:**
- - Manipulate game pieces effectively.
- - Score efficiently throughout the match.
- - Perform consistently under competition conditions.
- - Work efficiently with an alliance partner.
- Match strategy and performance should be **continuously evaluated** throughout the season to improve results.

Defining The Engineering Challenge

- Understanding the problem is the **first stage of the engineering design process.**
- **Clearly identifying challenges helps develop a robot that is:**
- - Purposeful
- - Efficient
- - Reliable
- - Strategically effective
- A well-defined problem leads to **better design decisions**, stronger mechanisms and improved competition performance.

Identifying Design Requirements

- The game **must be analysed** before any design decisions are made.
- **Identify:**
- - Scoring opportunities.
- - Field layout and obstacles.
- - How game pieces interact with the field.
- - Advantages and disadvantages of different robot strategies.
- - The role of alliance cooperation during matches.
- Basing engineering decisions on **evidence, testing and observations.**

Field Components

----------Identify----------

Goal

To identify and understand the field components and game elements that the robot must interact with during a match.

Cups (56 Total)

- 20 match loads
- - 10 **red alliance**
- - 10 **blue alliance**
- 36 preplaced
- - 24 start **grey side up**
- - 12 start **clear side up**

Pins (63 Total)

- **Red/Blue Pins (24 total)**
- - 4 preplaced at the start of the match
- - 20 match loads
- - 10 **red alliance**
- - 10 **blue alliance**
- **Yellow/Red Pins (28 total)**
- - 2 preloads
- - 20 match loads
- - 8 preplaced on the field
- **Yellow/Yellow Pins (19 total)**
- - 2 match loads
- 17 preplaced on the field

Goals (9 Total)

- **Alliance Goals (4)**
- - 2 red goals
- - 2 blue goals
- Neutral Goals (5)
- - 4 short goals
- - 1 tall goal

Other Field Elements

- **4 Toggles** (one centered on each field wall)
- **4 Loader** (adjacent to each driver station)

Scoring Opportunities

----------Identify----------

Goal To identify the highest-value scoring opportunities and determine which objectives should be prioritised during a match.

Yellow Pins

- **Worth 10 points each**
- **Highest value** scoring opportunity
- Neutral game element, making it highly contested
- Maximise points by ensure the corresponding toggle is set to our Alliance colour before the match ends

Alliance Pins

- **Worth 5 points each**
- Provides **reliable and consistent** scoring opportunity
- Faster to score depending on the robot's position and match strategy

Autonomous Period

- **Worth 12 points**
- Winning the Autonomous Bonus provides an **early advantage**
- Requires a reliable and consistent autonomous routine
- Can influence the overall match outcome

Midfield

- **Worth 8 points** at the end of the match per robot
- Valuable **endgame** scoring opportunity
- Can significantly increase the final score and determine close matches

Autonomous Bonus	12 Points
Each Scored Alliance Pin	5 Points
Each Scored Yellow Pin	10 Points
Each Robot In Midfield	8 Points

Constraints

----------Identify----------

Goal

To identify the factors that influence the robot's performance and must be considered throughout the design process.

Constraint	Justification
Match Time	The robot has a **limited time to score**, making a fast and efficient scoring cycle efficient.
Field Layout	The robot must **maneuver** around goals, toggles and other field elements without getting stuck or losing efficiency.
Game Pieces	The robot must be capable of manipulating **multiple types of game pieces** (pins and cups)
Alliance Cooperation	The robot must **work effectively** with an alliance partner without interfering with their scoring or movement.
Opponent Defense	The robot must continue to score efficiently despite contact and defensive strategies from opposing alliances.
Scoring Priorities	**High value** scoring opportunities should be prioritized while balancing consistency.
Reliability	The robot must perform **consistently** with minimal mechanical problems.
Driver Control	The robot must be **simple to operate**, allowing drivers to sore accurately under match pressure.
Limited Resources	Only the available VEX parts, motors and materials can be used.
Size Limit	The robot must remain within the **official VEX** size requirements throughout the competition.

Robot Goals

----------Identify----------

Goal

To establish clear design requirements that guide the development of a robot capable of consistently performing our match strategy.

Characteristic	Justification
Simple and Reliable	Create a robot with **minimal unnecessary mechanisms** to reduce potential points for failure and consistently **performs its intended functions** through every match.
Fast and Efficient	Achieve **quick scoring and intake cycles** with minimizing unnecessary movement mechanism operation to **maximise scoring opportunities** during a match.
Adaptable and Versatile	Design the robot capable of **performing multiple tasks** while allowing mechanisms and strategies to be modified.
Strong and Durable	Build a **rigid and durable robot** that can withstand **collisions, defensive play and repeated use** while reducing the risk of mechanical failure during competition.
High Scoring Capability	Design the robot to **consistently score** high value game elements and **control important scoring areas.**
Easy Maintenance	Motors, gears and other important components should be **easy to access** so repairs can be **completed quickly** between matches.
Stable and Maneuverable	Maintain a **low and balanced center of gravity** while providing sufficient **speed, traction and precise control** so the robot can move quickly without tipping.

Drivetrain Goals

----------Identify----------

Goal

To identify the characteristics of an effective drivetrain and determine the requirements for a reliable and competitive robot.

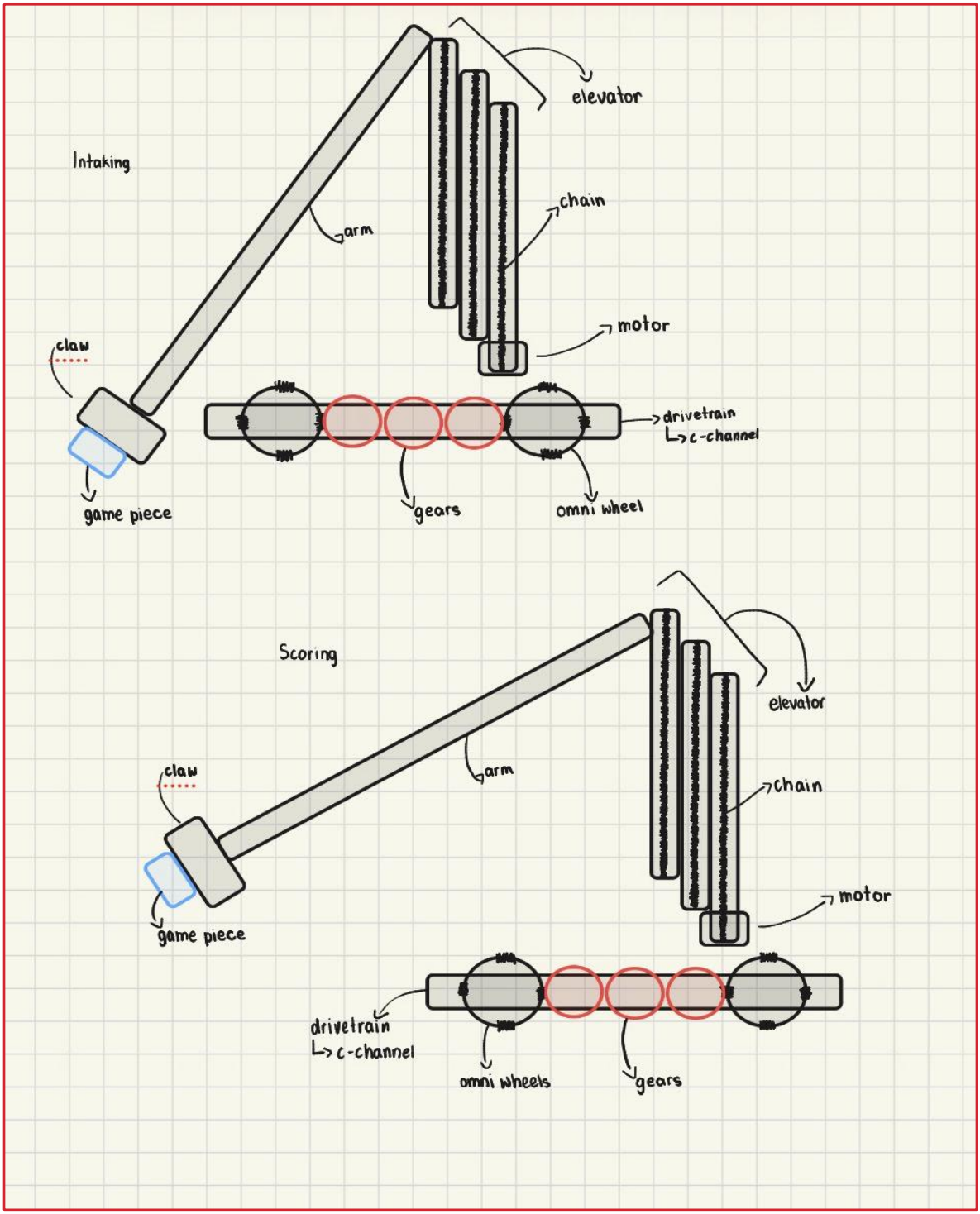
Characteristic	Justification
Strong and Durable	The robot must be **rigid** to support the robot's weight and withstand impacts. (Bearings, cross braces and proper construction)
Balanced Speed and Power	A suitable **gear ratio** provides a balance between speed and torque allowing efficient scoring while still being capable of defending.
Good Traction	High traction **prevents wheel slip**, improves **acceleration** and allows the robot to resist defensive pushing from opponents.
Compact and Accessible	A compact drivetrain leaves more room for other mechanisms, and **easy access** simplifies maintenance and allows damaged components to be replaced quickly.
Reliable	The drivetrain should **perform consistently** throughout every match, ensuring predictable movement and improving the accuracy of autonomous routines.
Easy Maintenance	Motors, gears and axles should be **easy to access** so repairs can be completed quickly between matches.
Stable	A stable drivetrain improves driver control, **reduces the amount of unwanted movement** and provides a solid base for lifting and scoring mechanisms.

BRAINSTORM

How can we solve the issue?

Robot Design 1

----------Design Sketch----------



Robot Design 1

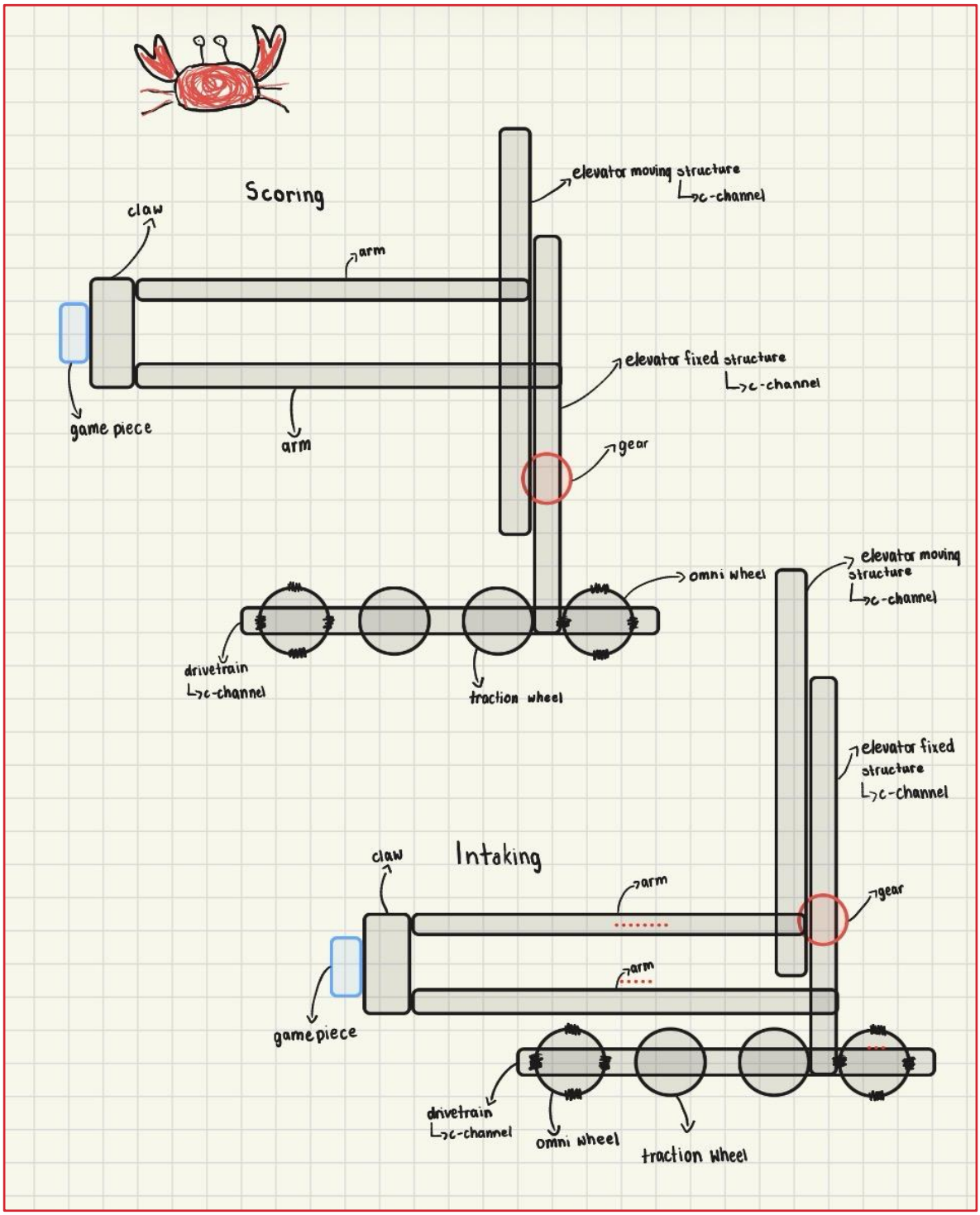
----------Analyze----------

Goal | Identify the strengths and weaknesses of the robot design

Strengths	Weaknesses
• **Vertical Expansion:** A double arm design enables great height to be reached. • **Large Reach:** The arm allows extension to game pieces that can be reached from a far distance. • **Compact Retraction:** The elevator can lower the arm which provides more stability when driving. • **Versatile Scoring:** A wide range of scoring positions and angles is available. • **Visibility:** Most mechanisms are mounted centrally which enables a relatively clear view of the field. • **Modular Design:** Main moving mechanisms can be tested and improved independently. • **Continued Functionality:** There is continued operation if one mechanism breaks.	• **Center Of Gravity:** High, resulting in greater chances of tipping when fully extended. • **Increased Weight:** Two separate mechanisms that don't depend on each other adds significant weight. • **Increased Toque:** It requires strong motors and gear reductions to lift game pieces reliability. • **Slower Cycle:** Each cycle to score is significantly longer including raising the elevator and positioning arm. • **Complex Construction:** Multiple moving mechanisms results in harder building and maintenance. • **Power Consumption:** Multiple moving mechanisms places high loads resulting in motor strain and battery consumption. • **Weight Distribution:** Must be optimised to ensure balance and prevent tipping.

Robot Design 2

----------Design Sketch----------



Robot Design 2

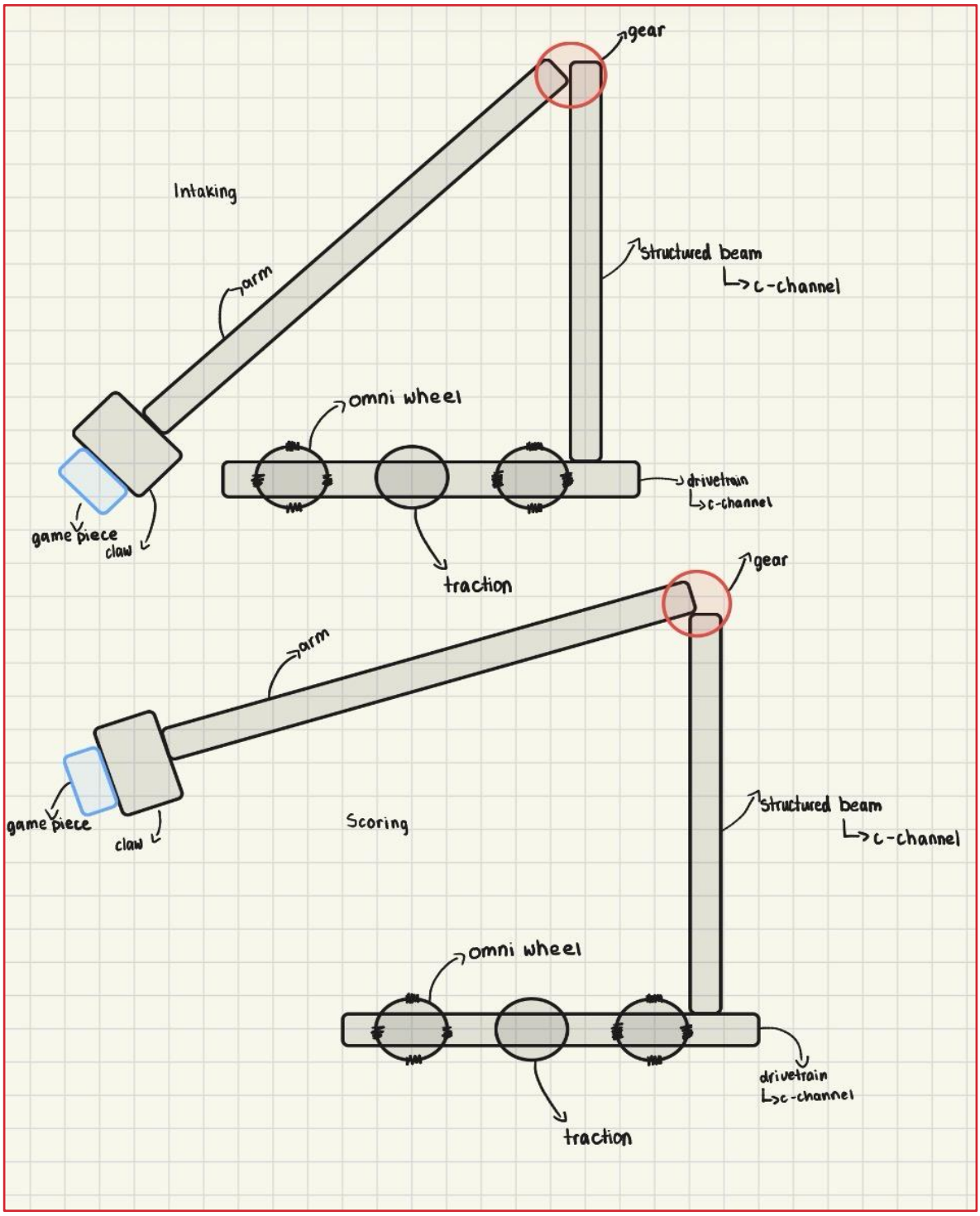
----------Analyze----------

Goal | Identify the strengths and weaknesses of the robot design

Strengths	Weaknesses
• **Simple Mechanism:** They are easier to build, maintain and troubleshoot with. • **Strong Support:** The dual arm design provides more support and improves the arm's stability. • **Accurate Scoring:** A leveled arm increases the scoring precision. • **Rigid Structure:** The elevator provides a strong and stable lifting mechanism for the arm. • **Reliable Operation:** With fewer moving mechanisms, there are less components that can fail. • **Compact design:** Staying within robot's frame improves maneuverability. • **Weight Distribution:** The elevator supports the arm's weight, reducing stress on a single pivot point.	• **Independent Arm Movement:** The angle cannot be adjusted separately from the elevator. • **Complete Dependence:** With a broken elevator, no other moving mechanism can function to score properly. • **Increased Torque:** High motor load where the elevator must lift the arm and the game piece. • **Slower Cycle:** Moving from an intake to scoring position results in slower scoring. • **Center Of Gravity:** High, resulting in greater chances of tipping when fully extended. • **Limited Flexibility:** The fixed angle creates difficult interactions with game pieces from different positions. • **Difficult Repairs:** Problems including the elevator would impact the entire scoring mechanism.

Robot Design 3

----------Design Sketch----------



Robot Design 3

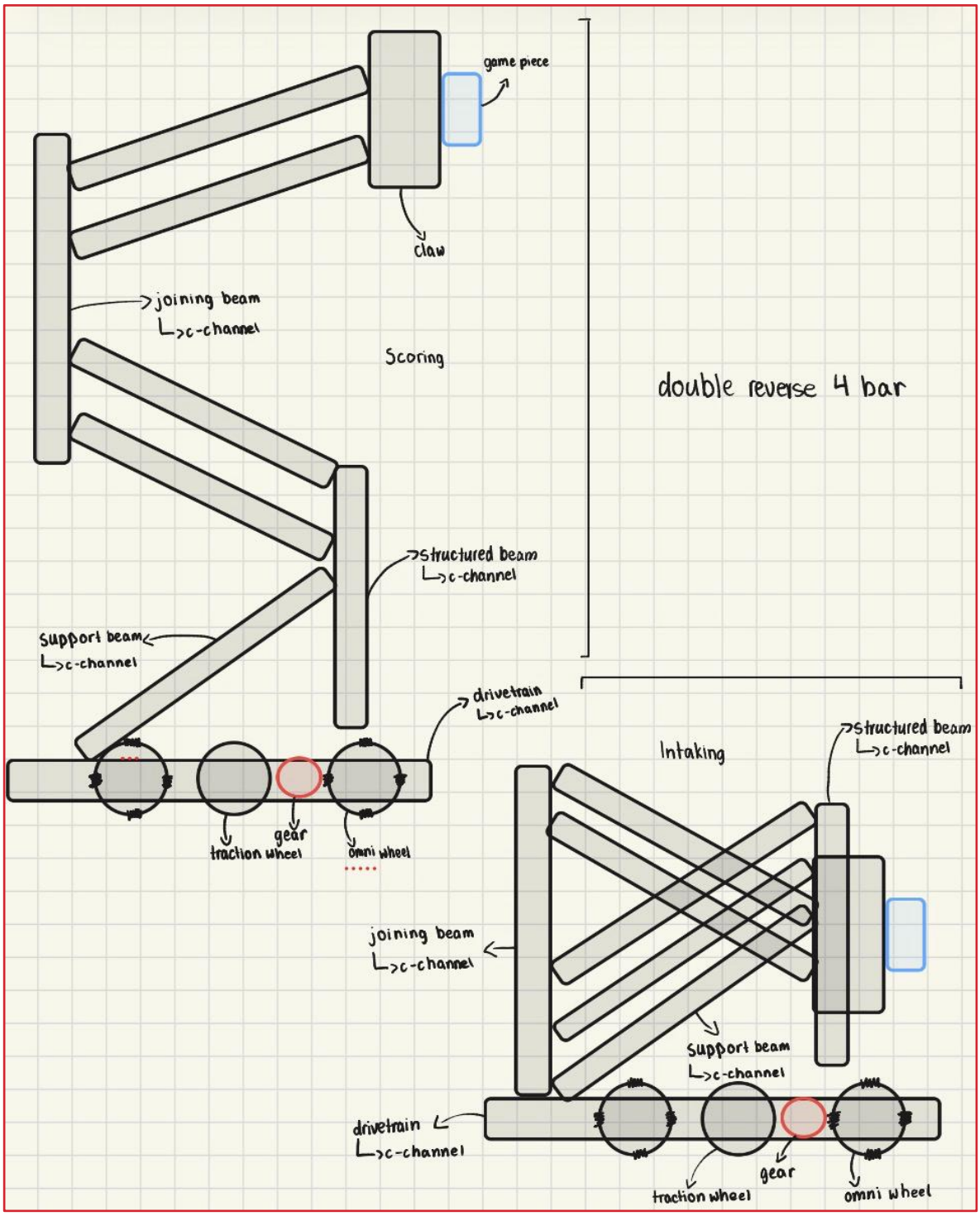
----------Analyze----------

Goal | Identify the strengths and weaknesses of the robot design

Strengths	Weaknesses
• **Simple Design:** One singular pivoting arm for intaking and scoring. • **Fast Cycle:** There is a direct pivot which reduces the cycle time it needs to score and intake. • **Lightweight:** There is less strain on the motors with less moving mechanisms. • **High Reliability:** There are fewer moving mechanisms which reduces the amount of mechanical issues. • **Quick Response:** There is rapid movement that comes from different positions of the robot. • **Strong Structure:** The fixed support beam provides a rigid mounting point for the arm. • **Power Consumption:** There are fewer moving mechanisms which reduces the battery consumption and less motor strain.	• **Scoring Height:** The height range that can be achieved is constricted within the robot's pivoting range. • **Increased Torque:** There is a large load on the pivot point which requires strong gearing and results in motor strain. • **Center Of Gravity:** High, resulting in greater chances of tipping when the robot is fully extended. • **Stopped Functionality:** If the pivot mechanism breaks, the entire arm becomes unfunctional and unusable. • **Precise Positioning:** Fine height adjustments are difficult to achieve without fixed positions. • **Reduced Versatility:** The fixed pivot point limits scoring from different heights and angles.

Robot Design 4

----------Design Sketch----------



Robot Design 4

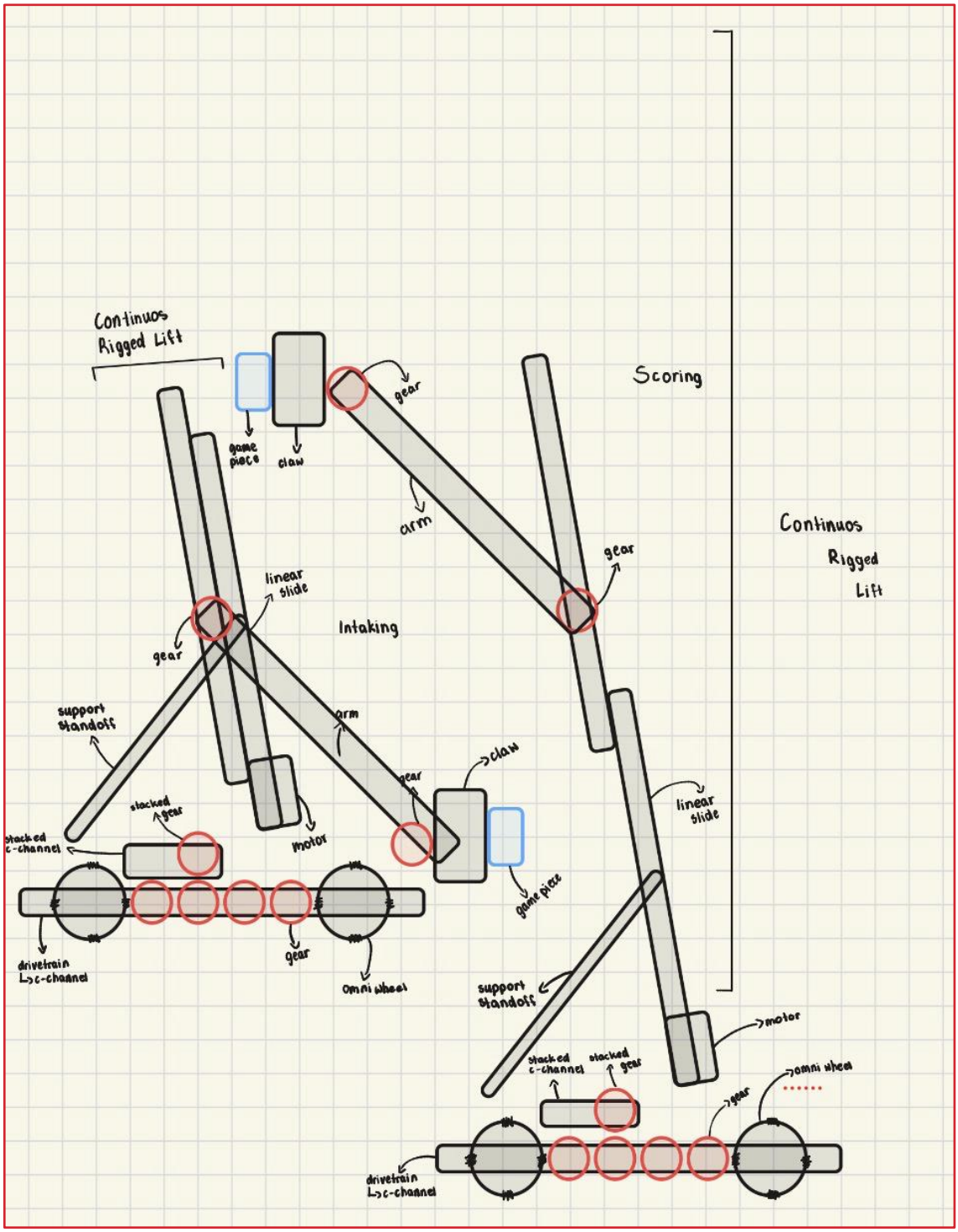
----------Analyze----------

Goal | Identify the strengths and weaknesses of the robot design

Strengths	Weaknesses
• **Vertical Expansion:** The double reverse 4 bar can reach high scoring locations. • **Compact Storage:** The linkage folds neatly within the robot, reducing its size during intaking and driving. • **Smooth motion:** The double reverse 4 bar provides a controlled and consistent path for the claw. • **Versatile Scoring:** A wide range of scoring positions and angles is available. • **Strong Lift:** The dual link design distributes forces, improving rigidity and reducing flex. • **Stable Handling:** The claw stays at a consistent orientation throughout lifting game pieces. • **Fast Cycle:** The linkage extends and retracts quickly, allowing efficient scoring.	• **Complex Design:** The double reverse 4 bar has many moving mechanisms, making it more difficult to build and tune. • **Precise Alignment:** Poorly aligned linkages can cause binding and reduce performance. • **Higher Maintenance:** More joints and bearings increase wear and require regular maintenance. • **Heavy Mechanism:** The additional beams and linkages increase the robot's overall weight. • **Increased Torque:** Lifting the linkage requires strong gearing and may result in motor strain. • **Stopped Functionality:** If the main linkage or gear breaks, the entire scoring system becomes unfunctional and unusable. • **Space:** The folded linkage occupies a significant amount of room inside the robot.

Robot Design 5

----------Design Sketch----------



Robot Design 5

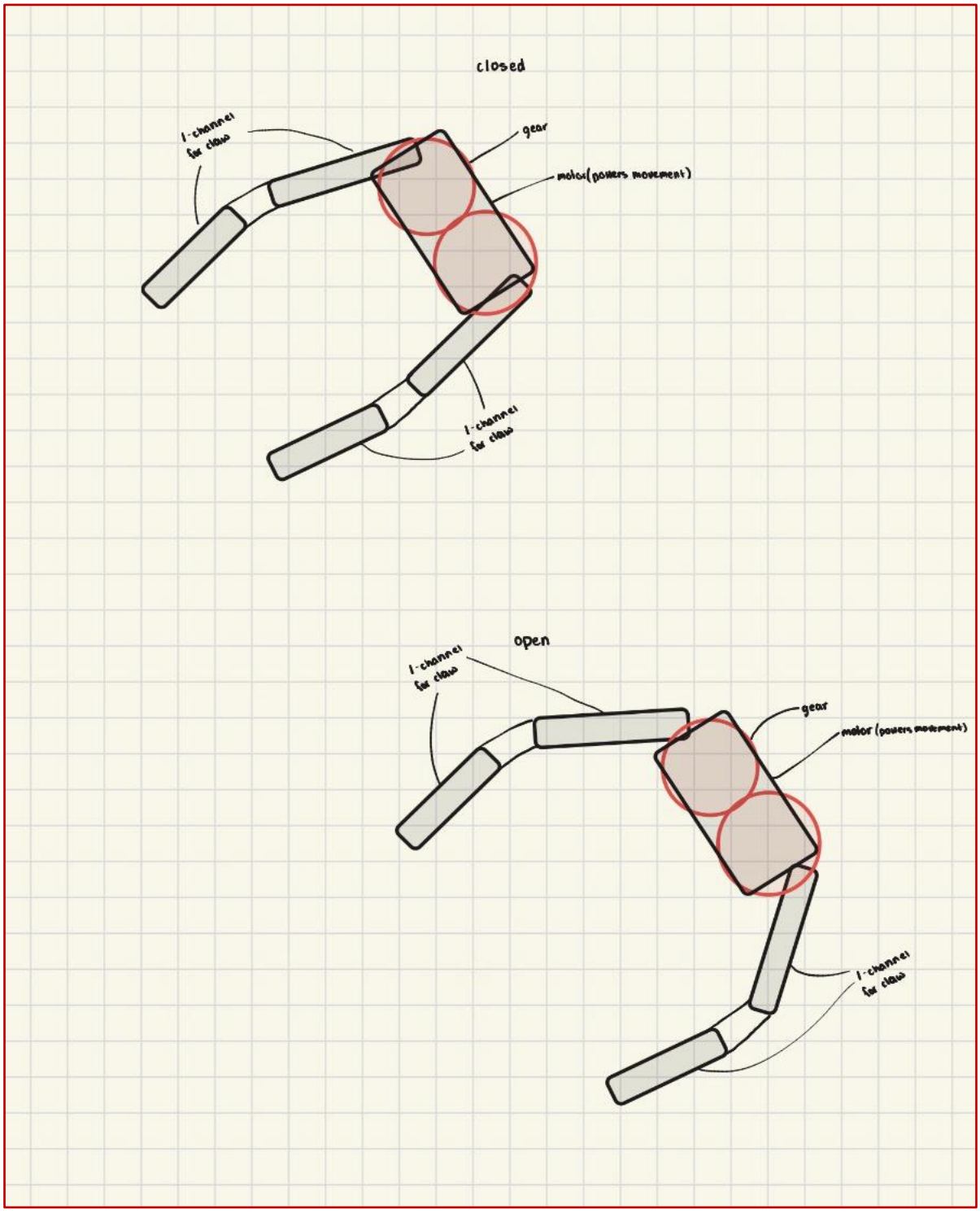
----------Analyze----------

Goal | Identify the strengths and weaknesses of the robot design

Strengths	Weaknesses
• **Vertical Expansion:** The continuous rigged lift can reach high scoring locations. • **Compact Storage:** The lift folds neatly within the robot, reducing its size during intaking and driving. • **Smooth motion:** The continuous rigged lift provides a controlled and consistent path for scoring. • **Versatile Scoring:** A wide range of scoring positions is available from the linear slides. • **Strong Lift:** Multiple support points distributes forces, improves rigidity and reduces stress on individual components. • **Stable Handling:** The rigging supports the arm and maintains its position when lifting. • **Fast Cycle:** The linear slides extend and retract quickly, allowing efficient scoring.	• **Complex Design:** The continuous rigged lift has many moving mechanisms, making it more difficult to build and tune. • **Precise Alignment:** The linkage slides must be aligned to prevent binding and reduce performance. • **Higher Maintenance:** Repairing or replacing the rigging or linear slides would take a long time. • **Heavy Mechanism:** The additional beams and linkages increase the robot's overall weight. • **Increased Torque:** The motors must lift the arm and the game piece at the same time. • **Stopped Functionality:** If the rigging becomes loose, tangled or breaks, the lift will stop functioning. • **Intake Flexibility:** The intake position is fixed, which makes intaking from different positions more difficult.

Claw Design 1

----------Design Sketch----------



Claw Design 1

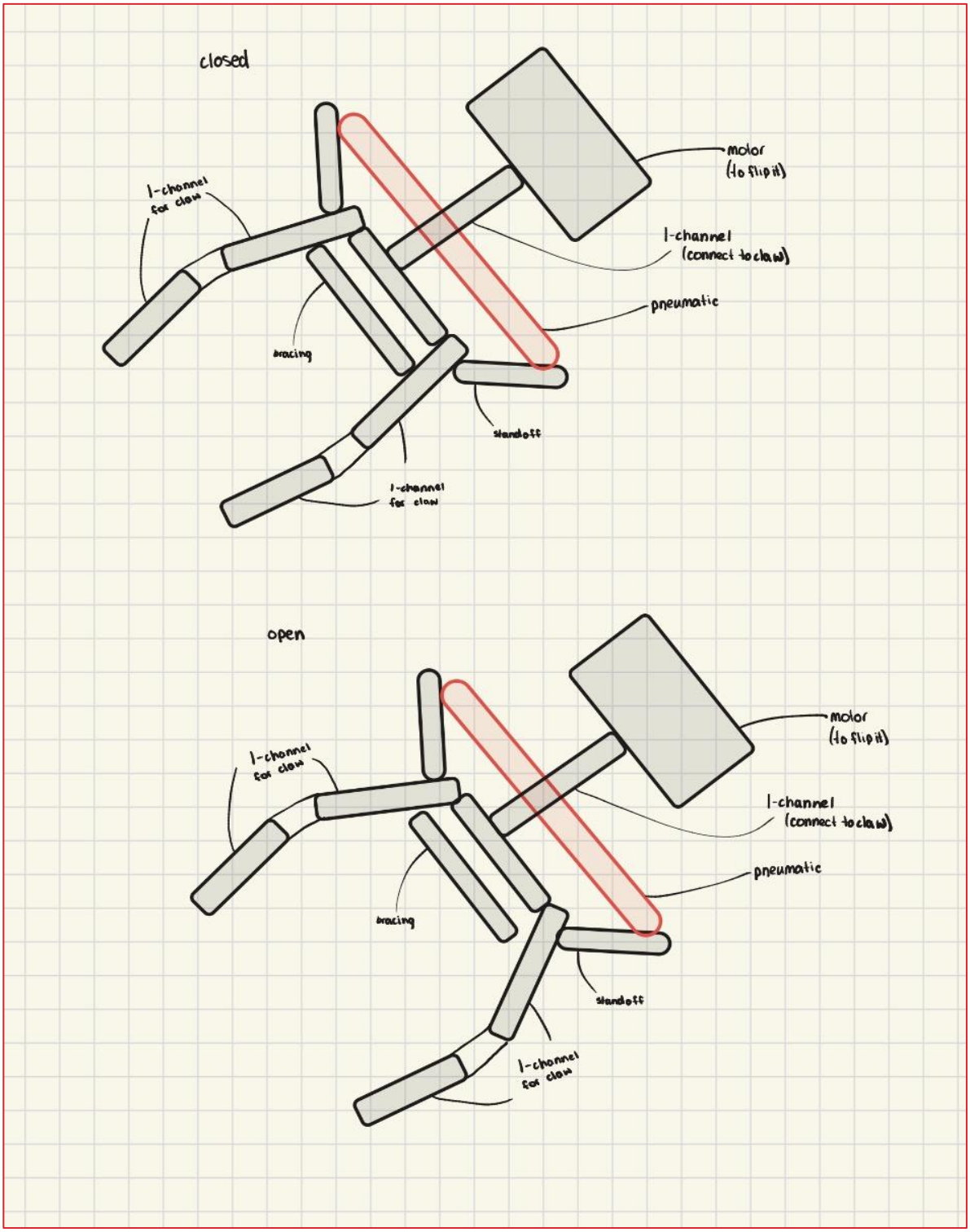
----------Analyze----------

Goal | Identify the strengths and weaknesses of a claw design

Strengths	Weaknesses
• **Simple And Reliable:** The mechanism uses a straightforward design with a motor transferring power through gears. • **Good Gripping Force:** Gear reduction can increase torque, allowing the claw to apply greater force when gripping game objects. • **Precise Control:** The motor can be programmed to open and close the claw by specific amounts, improving consistency. • **Compact Horizontal Design:** Mounting the motor above the claw can keep the mechanism compact and reduce the number of additional components. • **Easy To Construct And Maintain:** Compared with more complex mechanisms, a gear-driven claw is easier to build, troubleshoot, and repair.	• **Additional Weight:** Mounting the motor and gears above the claw can increase the weight of the mechanism. • **Limited Opening Range:** A traditional claw may only be able to open to a certain width. • **Potential Gear Slippage Or Wear:** If the claw experiences high forces, the gears may wear or skip if they are not properly supported or secured. • **Slower Movement With Gear Reduction:** Increasing torque through gear reduction may reduce the speed at which the claw opens and closes. • **Can Take Up Vertical Space:** The motor positioned above the claw may increase the overall height of the mechanism, which could interfere with other robot components or game requirements.

Claw Design 2

----------Design Sketch----------



Claw Design 2

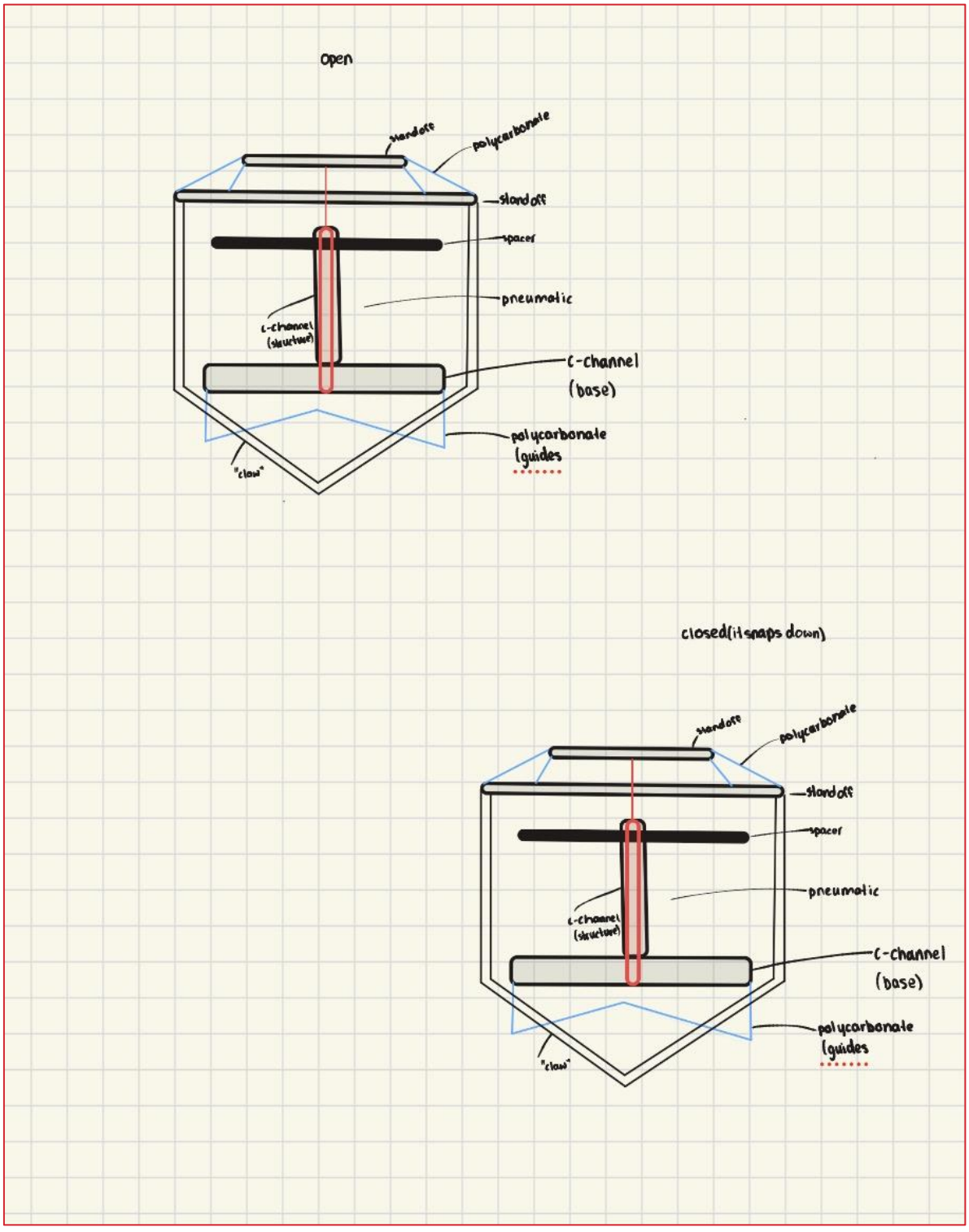
----------Analyze----------

Goal | Identify the strengths and weaknesses of a claw design

Strengths	Weaknesses
• **Greater Versatility:** The claw can collect and manipulate game objects, allowing it to perform multiple functions with one mechanism. • **Object Reorientation:** The flipping mechanism allows the robot to change the orientation of pins or cups. • **More Efficient:** Instead of requiring separate mechanisms to collect and reposition game objects, the claw may be able to perform both tasks. • **Improved Adaptability:** The ability to flip the claw could allow the robot to respond to game objects approaching from different orientations. • **Traditional Claw :** The basic claw design remains relatively simple and can provide controlled and consistent gripping.	• **Mechanical Complexity:** Adding a flipping mechanism requires additional motors, gears, or structural components, creating more potential points of failure. • **Additional Weight:** The rotating mechanism may increase the overall weight of the robot, particularly if the claw is mounted away from the drivetrain. • **Centre of Gravity:** If the claw is mounted high on the robot, the additional components could make the robot more prone to tipping. • **Programming:** The robot must accurately control both the claw's gripping movement and its flipping position, which could increase autonomous complexity. • **Space Constraints:** The claw needs sufficient clearance to rotate without colliding with other parts of the robot.

Claw Design 3

----------Design Sketch----------



Claw Design 3

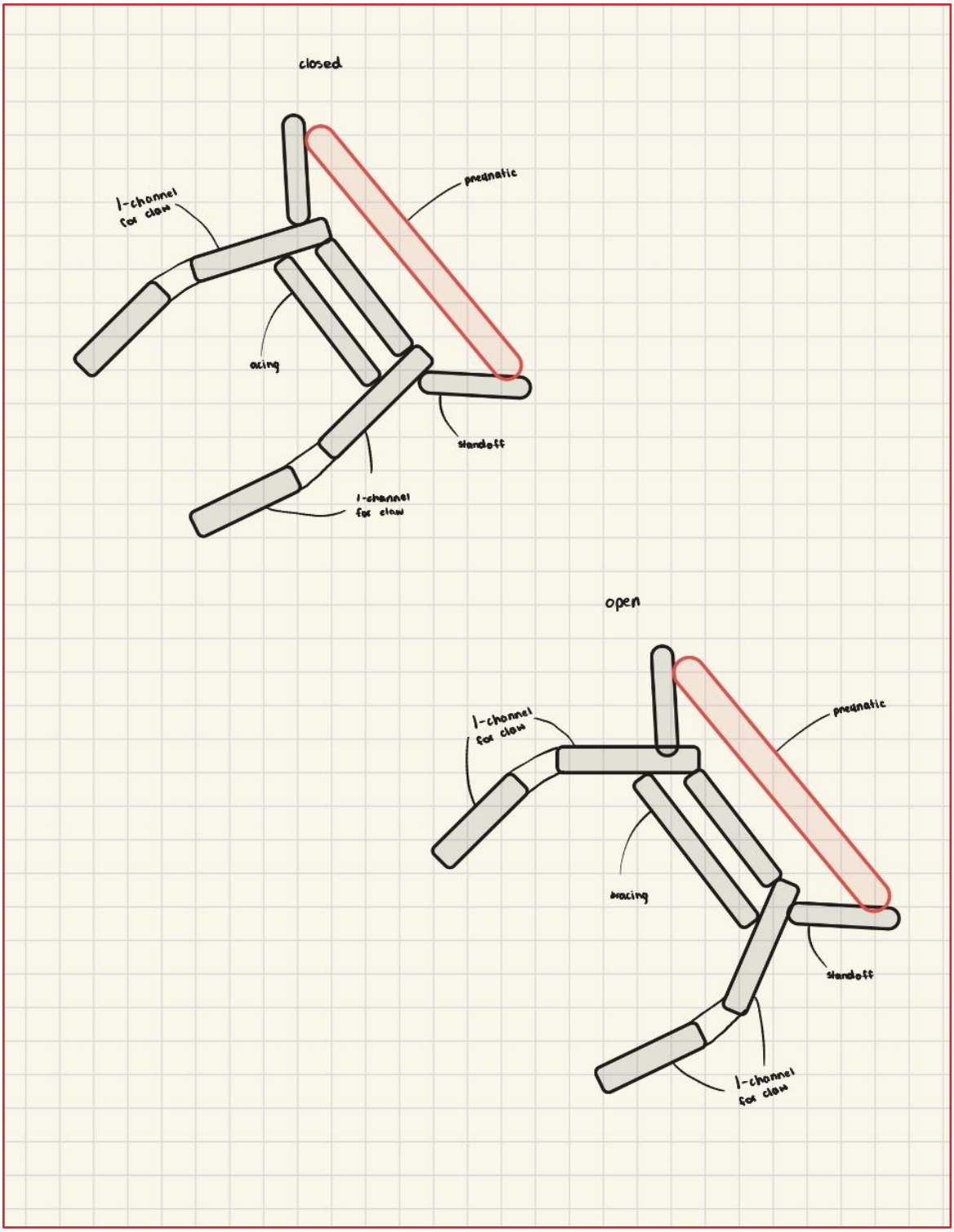
----------Analyze----------

Goal | Identify the strengths and weaknesses of a claw design

Strengths	Weaknesses
• **Collection Area:** The wide funnel opening provides a large target area, making it easier to collect game objects without requiring extremely precise alignment. • **Guides Objects Into Position:** The funnel shape naturally directs objects towards the centre of the mechanism, improving consistency of collection. • **Versatile Object Handling:** The motorised arms can provide additional control over objects, allowing the mechanism to assist with collecting, positioning, or manipulating pins and cups. • **Faster Collection:** Because the intake has a wide opening, the robot may be able to approach objects from a wider range of angles, reducing the time required for precise positioning.	• **Space Required:** The funnel and supporting structure take up a significant amount of space, which could limit room for other mechanisms. • **Additional Weight:** The large structural components, combined with the motorised arms, add weight to the robot. • **Complex Construction:** The mechanism requires multiple structural supports and carefully positioned components to ensure that the funnel, arms, and motors operate without interference. • **Risk Of Objects Becoming Stuck:** Depending on the exact dimensions and movement of the mechanism, cups or pins could potentially become jammed inside the funnel.

Claw Design 4

----------Design Sketch----------



Claw Design 4

----------Analyze----------

Goal | Identify the strengths and weaknesses of a claw design

Strengths	Weaknesses
• **Large Intake Area:** The wide opening makes it easier to collect cups and other game objects without requiring extremely precise alignment. • **Efficient Object Collection:** The funnel shape helps guide game objects towards the centre of the mechanism, improving the consistency of collection. • systems. • **Strong Autonomous:** The large intake area provides a greater margin for error, which could improve the reliability of autonomous movements. • **Adaptable To Different Situations:** The ability to collect and reorient objects gives the robot greater flexibility when responding to different match and scoring situations.	• **Requires Careful Adjustment:** The mechanism will need to be tested and adjusted to ensure that the movement is smooth and consistent. functionality. • **Requires Sufficient Clearance:** The mechanism will need enough space to move through its full range of motion without interfering with other components. • **Additional Weight:** The structural components and flipping mechanism may add some weight to the robot, which will need to be considered when balancing the overall design. • **Pneumatic Dependence:** The flipping mechanism relies on the pneumatic system, meaning it will need to be designed to ensure consistent and reliable operation.

4 Wheel Tank Drive

----------Analyze----------

Goal Identify the strengths and weaknesses of a 4 wheel tank drive.

Strengths	Weaknesses
• **Simple Construction:** The drivetrain uses fewer components, making it easier and faster to build, repair and modify. • **Motor Efficiency:** Fewer motors are required for the drivetrain, allowing additional motors to be used for mechanisms such as the intake or elevator. • **Compact Design:** The smaller drivetrain can leave more space within the robot for other mechanisms. • **Easy Programming:** Tank drive is relatively simple to code and control, reducing programming complexity. • **Lower Weight:** Using fewer wheels and components can reduce the overall weight of the robot.	• **Limited Traction:** With only four wheels in contact with the field, the robot may have less pushing power than a six-wheel drivetrain. • **Reduced Stability:** The drivetrain provides less support for a larger or heavier robot. • **Easier to Push:** Lower traction may make the robot more vulnerable when interacting with opposing robots. • **Less Weight Distribution:** The robot's weight is distributed across fewer wheels, which may place greater pressure on each wheel. • **Turning Friction:** Depending on the wheel type, turning can create friction and reduce efficiency.

6 Wheel Tank Drive

----------Analyze----------

Goal Identify the strengths and weaknesses of a 6 wheel tank drive.

Strengths	Weaknesses
• **Greater Traction:** The additional wheels increase the contact area with the field, providing stronger pushing power. • **Improved Stability:** Six wheels provide a wider and more supported base, which is beneficial for robots with taller mechanisms. • **Better Weight Distribution:** The robot's weight is spread across more wheels, reducing pressure on individual wheels. • **High Durability:** The drivetrain has multiple points of contact with the field, creating a strong and reliable base. • **Strong Defensive Ability:** Greater traction can make it more difficult for other robots to push the robot away from its intended position.	• **Difficult Turning:** Increased traction can create greater friction, making the robot harder to turn smoothly. • **Additional Weight:** More wheels and structural components can increase the overall weight of the robot. • **Greater Complexity:** The drivetrain requires more components and careful construction. • **Reduced Internal Space:** The larger drivetrain may take up valuable room needed for other mechanisms. • **Higher Friction:** Scrubbing during turns can reduce efficiency and place additional strain on the motors.

X – Drive

----------Analyze----------

Goal | Identify the strengths and weaknesses of a x drive.

Strengths	Weaknesses
• **Holonomic Movement:** The robot can move forwards, backwards, sideways and diagonally without rotating. • **Precise Positioning:** Sideways movement allows the robot to make small adjustments when approaching scoring areas. • **High Manoeuvrability:** The robot can quickly change its direction of movement. • **Flexible Scoring:** The robot can approach game elements from multiple angles without repositioning its entire drivetrain. • **Efficient Movement:** Diagonal movement can provide more direct pathways around the field.	• **Complex Construction:** The wheel and motor arrangement is more difficult to build correctly. • **More Difficult Programming:** Holonomic movement requires more complex programming and driver control. • **Lower Pushing Power:** Omni wheels generally provide less traction than standard traction wheels. • **Greater Driver Skill Required:** The additional movement directions can make the drivetrain harder to control effectively. • **Space Requirements:** The X-shaped motor and wheel configuration may limit the available space for other mechanisms.

SELECT

Choose a design which best fits our problem and offers the best solution.

Criteria

----------Select----------

Goal

To compare multiple robot concepts against a defined success criteria and select the design that best meets our strategic, mechanical and competition requirements.

Criteria	What We Require
Scoring Capability	Ability to reach and score at **different heights** and **effectively interact** with important game pieces.
Reliability	Ability to **perform consistently** with **minimal risk of mechanical failure** during competition.
Speed And Efficiency	Ability to **complete scoring cycles quickly** while **minimizing unnecessary movement.**
Strength And Durability	Ability to **withstand collisions, defensive play and repeated operation** without bending or breaking.
Stability	Ability to maintain a **balanced center of gravity** and avoid tipping when mechanisms are extended.
Versatility	Ability to **perform multiple tasks** and **adapt** to different match situations.
Maintenance	The **ease of accessing, repairing ad replacing** components of the robot between matches.
Complexity	How difficult the robot design is to **build, program, operate and maintain.**
Compactness	How efficiently the mechanisms **use available space** without interfering with the drivetrain or other mechanisms.

Robot Ideas

----------Select----------

Goal

To objectively compare each robot design against the same success criteria and identify the strengths, weaknesses and overall performance of each concept.

Criteria	D1	D2	D3	D4	D5
Scoring Capability	9	7	6	9	10
Reliability	7	8	8	7	9
Speed And Efficiency	7	6	9	8	9
Strength And Durability	7	9	7	8	9
Stability	6	7	6	8	9
Versatility	9	6	6	9	10
Maintenance	6	8	9	6	7
Complexity	6	8	9	5	6
Compactness	7	7	9	7	8
Overall Score	64/90	66/90	69/90	67/90	77/90

Ranking

1. Design 5 – 77/90
2. Design 3 – 69/90
3. Design 4 – 67/90
4. Design 2 – 66/90
5. Design 1 – 64/90

Design 5 achieved the highest overall score because it provides the best balance between **scoring capability, reliability, stability, versatility and efficiency.**

Design Selection

----------Reasoning----------

Goal

To identify the characteristics of an effective drivetrain and determine the requirements for a reliable and competitive robot.

Decision
Design 5 was selected because it best satisfies our criteria for success and provides the strongest overall balance between scoring capability, reliability, speed, strength, stability and versatility. While its complexity presents additional engineering challenges, the potential competitive advantages outweigh these limitations.

Why D5?	Justification
High Scoring Capability	The large vertical range allows the robot to reach different scoring positions and provides greater scoring flexibility.
Versatility	The robot can perform a wider range of tasks without requiring separate mechanisms.
Stable Structure	The supported lift provides a strong framework while maintaining stability.
Efficient Scoring	The robot can quickly transition from different positions, supporting fast scoring cycles.
Reliable	The design can repeatedly move through its range of motion, making it suitable for consistent competition use.
Competitive Advantage	Its combination of reach, versatility and reliability gives us more strategic options during matches.

Design Selection

----------Reasoning----------

Goal

To identify the characteristics of an effective drivetrain and determine the requirements for a reliable and competitive robot.

Advantages	Disadvantages
Faster acceleration	The large vertical range allows the robot to reach different scoring positions and provides greater scoring flexibility.
Versatility	The robot can perform a wider range of tasks without requiring separate mechanisms.
Stable Structure	The supported lift provides a strong framework while maintaining stability.
Efficient Scoring	The robot can quickly transition from different positions, supporting fast scoring cycles.
Reliable	The design can repeatedly move through its range of motion, making it suitable for consistent competition use.
Competitive Advantage	Its combination of reach, versatility and reliability gives us more strategic options during matches.

Wheel Selection

----------Select----------

Goal

To compare the advantages and disadvantages of 2.75-inch wheels to determine whether they are the most suitable option for our robot.

Advantages	Justification
Higher Top Speed	Larger wheels have a greater circumference, allowing the robot to travel across the field more quickly.
Greater Ground Clearance	The increased wheel diameter provides better ground clearance, which is important for crossing the park barrier.
Greater Pushing Capability	The larger wheel radius provides greater mechanical leverage, which can improve the robot's ability to push against other robots.

Disadvantages	Justification
Slower Acceleration	Larger wheels require more torque to accelerate, resulting in slightly slower initial acceleration.
Higher Centre Of Gravity	Larger wheels raise the robot's overall height, increasing the risk of tipping if the weight distribution is not properly balanced.
Reduced Turning Precision	Each wheel rotation moves the robot a greater distance, making small adjustments and precise autonomous movements more difficult.

Wheel Selection

----------Select----------

Goal

To compare the advantages and disadvantages of 3.25-inch wheels to determine whether they are the most suitable option for our robot.

Advantages	Justification
Faster Acceleration	Smaller wheels require less torque to rotate, allowing the robot to accelerate more quickly.
Lower Centre of Gravity	The smaller wheel size helps lower the robot's centre of gravity, reducing the likelihood of tipping.
Greater Precision	Smaller wheels provide more precise movement and control, particularly beneficial during autonomous.

Disadvantages	Justification
Lower top speed	Smaller wheels have a shorter circumference, meaning the robot travels a shorter distance with each rotation.
Lower Ground Clearance	The reduced wheel diameter decreases the robot's ground clearance, making it more difficult to travel over obstacles.
Increased motor workload	The wheels must rotate more times to travel the same distance, placing greater demand on the motors and potentially increasing heat generation and wear.

Wheel Selection

----------Reasoning----------

Goal

To explain the importance for selecting the correct wheel size and justify our final decision to use 2.75-inch wheels for our first robot.

Importance

Choosing the correct wheel size is an essential part of designing our drivetrain, as it directly impacts the robot's **speed, torque, acceleration, centre of gravity and overall stability**. Larger wheels can allow the robot to travel further with each rotation, potentially increasing its top speed, while smaller wheels can provide greater control and allow for more precise movements. Therefore, it is important to select a wheel size that provides the best balance for our robot's overall design and game strategy.

Decision

We have decided to use **2.75-inch wheels for our first robot** because we are prioritising a reliable and precise autonomous. The smaller wheel size allows for finer adjustments and more controlled movements, which we believe will improve the robot's ability to accurately position itself when approaching game objects and specific field locations. Additionally, the 2.75-inch wheels contribute to a **lower centre of gravity**, helping to improve the robot's overall stability during acceleration, turning and interaction with other robots. They also complement our chosen **450 RPM drivetrain**, creating a balance between speed, acceleration, torque and control.

Gear Selection

----------Select----------

Goal

To compare the advantages and disadvantages of 450 rpm gears to determine whether they are the most suitable option for our robot's drivetrain.

Advantages	Justification
Greater Torque	It provides slightly more torque making it more effective when pushing against other robots or overcoming defensive pressure.
Faster Acceleration	The increased torque allows the robot to accelerate more quickly, which is beneficial for reaching goals efficiently and responding to defensive plays.
Autonomous Control	The lower speed provides greater control during autonomous, reducing the likelihood of overshooting target positions and improving movement accuracy.

Disadvantages	Justification
Lower Top Speed	The reduced maximum speed means the robot may take longer to travel across the field.
Potential Autonomous Limitations	Although the drivetrain offers greater control, its slower top speed may limit the number of tasks the robot can complete during the autonomous period.

Gear Selection

----------Select----------

Goal

To compare the advantages and disadvantages of 450 rpm gears to determine whether they are the most suitable option for our robot's drivetrain.

Advantages	Justification
Higher Cycle Efficiency	The higher speed allows the robot to cover more distance per second, improving its ability to move quickly and complete scoring cycles more efficiently.
Higher Top Speed	The increased maximum speed allows the robot to travel across the field more quickly. This could be particularly beneficial during autonomous, as the robot may be able to complete more tasks within the limited time available.

Disadvantages	Justification
Lower Torque	It produces less torque, making it less effective for pushing against opposing robots or responding to defensive plays.
Slower Acceleration	The drivetrain may accelerate more slowly, making it less effective when travelling short distances or when rapid changes in direction and speed are required.

Gear Selection

----------Reasoning----------

Goal

To explain the importance for selecting the correct gear size and justify our final decision to use 450 rpm gears for our first robot's drivetrain.

Importance

Choosing the gearing for our drivetrain is a very important decision, as it directly influences both the **speed** of the robot and the **torque** it can produce. We need to ensure that there is an effective balance between these two factors, as prioritising one too heavily could negatively affect the robot's overall performance. After comparing several options, we narrowed our decision down to **450 RPM and 480 RPM**, as we believed these two options provided the best overall capabilities for our robot and game strategy.

Decision

We have decided to use a **450 RPM drivetrain** because we believe it provides the most **suitable balance of speed, acceleration and torque.** One of the main advantages of this gearing is its **stronger acceleration**, allowing the robot to reach its desired speed more quickly. This is particularly beneficial when travelling shorter distances, quickly reaching game objects or responding to changes during a match.
The 450 RPM drivetrain also provides slightly more torque than the 480 RPM option. This gives the robot **greater pushing power**, making it more effective when dealing with defensive pressure from opposing robots while still maintaining sufficient speed for offensive plays.

Elevator Design Problem

----------Identify----------

Goal

Design a lift that raises Pins and Cups to useful Goal heights while staying compact, stable and controllable.

The Match Need

- Raise the held game piece to several useful scoring heights.
- Keep the scoring path predictable for the driver.
- Retract low enough for driving, intaking and Endgame positioning.

The Design Problem

- A fixed-height mechanism cannot serve every useful stack level.
- More lift height adds moving mass, guide friction and tipping leverage.
- The elevator must share the robot with the intake, drivetrain, wiring and game pieces.
- The mechanism must lift, hold, place and return reliably.

What Success Looks Like

The robot can reach the planned scoring positions, hold the game piece during motion, and return the elevator without blocking another subsystem.

Elevator Requirements and Constraints

----------Identify----------

Goal

Turn the game task and the rest of the robot layout into practical elevator requirements.

Game Constraints

- Start inside 18 x 18 x 18 in.
- Stay inside 24 x 24 in horizontal expansion.
- Stay below 50 in before Endgame.
- Return to 18 in or lower when occupying Midfield during Endgame.

Engineering Requirements

- Compact storage and sufficient vertical travel.
- Repeatable scoring path with limited raised mass.
- No interference with intake, wiring or game-piece flow.
- Guided motion with manageable slack, flex and wear.
- Accessible tensioning, hard stops and replaceable wear parts.

CAD and Test Targets

- Required travel between the lowest and highest scoring positions.
- End-effector and game-piece load.
- Space remaining after the intake and drivetrain are placed.
- Motor setup, gearing and target lift time.
- Full-path clearance and dependable return.

Decision Rule

Reject any concept that cannot meet size, reach or path requirements. Rank the remaining concepts by stability, packaging, reliability, buildability and serviceability.

Weighted Decision Method

----------Identify----------

Goal

Use a transparent scoring method so the selected concept follows from the requirements instead of preference alone.

Rating Scale

1 = POOR FIT
2 = WEAK FIT
3 = WORKABLE WITH COMPROMISES
4 = STRONG FIT
5 = EXCELLENT FIT

WEIGHTED POINTS = (RATING / 5) x WEIGHT

Criterion	Wt	Reason for weight
Compact storage	20	Starting size can eliminate a concept
Vertical reach	20	Must reach intended scoring positions
Predictable path	15	Supports repeatable alignment
Stability	15	Limits raised mass and offset
Packaging	10	Protects space for other subsystems
Reliability	10	Controls slack, flex and wear
Buildability	5	Fits available parts and time
Serviceability	5	Keeps failure points accessible

Important Limitation

These ratings are pre-build engineering estimates. CAD and prototype evidence must later confirm or replace them.

DR4B - Operation and Evidence

----------Brainstorm----------

Goal

Determine whether a double reverse four-bar can provide long, controlled lift travel inside 4613R's packaging limits.

How It Works

- Two mirrored four-bar stages multiply reach.
- Parallel link geometry can approximate vertical motion.
- Cross-linking both sides helps synchronize the stages.
- Pivot locations and link lengths define the path.

Evidence and Implications

- Team 33 demonstrated a cross-linked DR4B at FRC scale.
- Long reach is possible without telescoping rails.
- NASA identifies high part count, mass, space use and sway as common drawbacks.
- 4613R would need rigid links and synchronized sides.



FRC Team 33 DR4B robot



Team 33 mechanism overview

DR4B - 4613R Evaluation

----------Brainstorm----------

Goal

Score the DR4B against the same criteria used for every concept.

Criterion	Wt	Rate	Predicted rationale
Compact storage	20	3	Folds, but long links still occupy volume
Vertical reach	20	5	Two stages provide excellent reach
Predictable path	15	4	Near vertical if geometry stays constrained
Stability	15	2	Many moving links; sway and high mass risk
Packaging	10	1	Large swept volume around the robot
Reliability	10	2	Backlash and synchronization at many joints
Buildability	5	2	High part count and alignment demand
Serviceability	5	2	Many pivots and fasteners to inspect

Result	4613R Judgement
60 / 100 Rank: 3rd	Excellent reach, but the swept volume, moving mass and number of joints create too much predicted integration risk for our compact concept.

Two-Bar Arm - Operation and Evidence

----------Brainstorm----------

Goal

Determine whether a simple pivoting arm can reach the scoring positions without unacceptable path and stability penalties.

How It Works

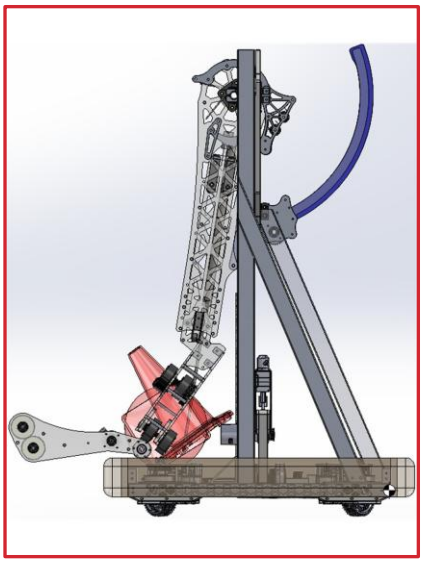
- A geared motor rotates the arm around one fixed pivot.
- A second member can brace the arm or control orientation.
- The end effector follows a circular arc.
- Counterbalancing reduces holding torque but adds hardware.

Evidence and Implications

- FRC 6560 used a rack-and-pinion pivot with an 80 degree range.
- Its design included an encoder, limit switch and constant-force springs.
- One joint is easy to sense and repair.
- 4613R would still need to manage arc motion, swing clearance and changing torque.



FRC 6560 pivot-arm robot



FRC 6560 pivot-arm CAD

Two-Bar Arm - 4613R Evaluation

----------Brainstorm----------

Goal

Score the pivoting arm against the same criteria used for every concept.

Criterion	Wt	Rate	Predicted rationale
Compact storage	20	3	Can fold, but arm length controls stored size
Vertical reach	20	3	Reach is limited by arm length and pivot
Predictable path	15	2	Height and horizontal reach change together
Stability	15	2	Load moves far from the drivetrain
Packaging	10	2	Needs swing clearance through the robot
Reliability	10	4	Few joints and direct sensing
Buildability	5	5	Lowest part count and simplest structure
Serviceability	5	5	Pivot, motor and sensor remain accessible

Result	4613R Judgement
58 / 100 Rank: 4th	The simplest option, but its arc path and horizontal mass offset are poor matches for repeatable vertical placement in a restricted envelope.

Cascade Elevator - Operation and Evidence

----------Brainstorm----------

Goal

Determine whether simultaneous stage motion gives the best balance of reach, path control and packaging.

How It Works

- Nested stages run on guides or bearing blocks.
- Every stage moves the same distance relative to its parent.
- All stages extend together.
- Carriage travel is the sum of simultaneous stage movement.

Evidence and Implications

- FRCDesign documents equal relative stage movement.
- Stage position is determinate at intermediate heights.
- Team 8033 used a three-stage WCP cascading elevator mounted at 41 degrees.
- Raising every stage together increases mid-extension centre of mass.



FRC Team 8033 elevator build



FRC Team 8033 cascade elevator

Cascade Elevator - 4613R Evaluation

----------Brainstorm----------

Goal

Score cascade rigging against the same criteria used for every concept.

Criterion	Wt	Rate	Predicted rationale
Compact storage	20	5	Nested stages retract into a short package
Vertical reach	20	5	Multiple stages provide excellent travel
Predictable path	15	5	Straight path and determinate stage positions
Stability	15	3	Every stage rises at partial height
Packaging	10	4	Narrow envelope but more rigging segments
Reliability	10	3	Multiple synchronized lines and guides
Buildability	5	3	Documented parts, but careful alignment needed
Serviceability	5	3	Several tension and attachment points

Result	4613R Judgement
84 / 100 Rank: 2nd	A strong, controllable option. It loses to continuous rigging because it raises more stage structure than required at partial height.

Continuous Elevator - Operation and Evidence

----------Brainstorm----------

Goal

Determine whether sequential stage motion best combines compact storage, long travel and partial-height stability.

How It Works

- Nested stages run on aligned linear guides.
- One rigging path transfers motion through the stages.
- Stages move sequentially according to friction and required force.
- The carriage follows a straight vertical path.

Evidence and Implications

- FRCDesign contrasts sequential motion with cascade's simultaneous motion.
- More stages remain nested at partial extension.
- Team 3255 documented continuous string rigging on its 2019 robot.
- Stage order, slack and line retention require deliberate design.



FRC Team 3255 continuous elevator



Continuous elevator scoring example

Continuous Elevator - 4613R Evaluation

----------Brainstorm----------

Goal

Score continuous rigging against the same criteria used for every concept.

Criterion	Wt	Rate	Predicted rationale
Compact storage	20	5	Nested stages retract into a short package
Vertical reach	20	5	Multiple stages provide excellent travel
Predictable path	15	5	Carriage follows a straight vertical path
Stability	15	4	More stages remain nested at partial height
Packaging	10	5	Narrow envelope with no pivot sweep
Reliability	10	3	Slack, wear and stage order need control
Buildability	5	3	Careful guidance and line routing required
Serviceability	5	4	Tension points can be designed for access

Result	4613R Judgement
90 / 100 Rank: 1st	The strongest predicted overall fit, provided rigging, guidance and stage-order risks are engineered before construction.

Weighted Concept Results

----------Select----------

Goal

Compare every concept using the same eight weighted requirements and calculate an overall pre-build ranking.

Criterion	Wt	DR4B	2-Bar	Cascade	Continuous
Compact storage	20	3	3	5	5
Vertical reach	20	5	3	5	5
Predictable path	15	4	2	5	5
Stability	15	2	2	3	4
Packaging	10	1	2	4	5
Reliability	10	2	4	3	3
Buildability	5	2	5	3	3
Serviceability	5	2	5	3	4
Weighted total	100	60	58	84	90
Rank	-	3rd	4th	2nd	1st

Ranking	Meaning
1. Continuous - 90/100 2. Cascade - 84/100 3. DR4B - 60/100 4. Two-bar - 58/100	The score selects a direction for CAD. It is not measured proof that the winning mechanism will work.

Why Continuous Won

----------Select----------

Goal

Explain why continuous rigging is the best first build direction and what could still change the choice.

What the Result Shows

- Continuous and cascade both scored strongly on reach and path control.
- Continuous kept more structure nested at partial height.
- It avoided the large swept path of a pivot arm.
- The ranking selected what to prototype first; it did not prove the final design.

Why Cascade Stayed as Backup

Cascade remained a strong option because its stages move in a predictable relationship. It would become preferable if continuous rigging repeatedly lost tension or became difficult to repair.

What Could Change the Decision

- A shorter required stroke could make a pivot arm practical.
- A heavier end effector would increase the stability penalty.
- CAD interference could eliminate a concept.
- Repeated rigging failures would move cascade ahead.
- The first build was a prototype, not a final answer.

Selected Direction - Continuous Elevator

----------Select----------

Goal

Record why continuous rigging was chosen and turn the decision into the first practical build.

Why We Selected It

- Highest weighted result: 90/100.
- Long travel from a compact nested package.
- Straight vertical scoring path.
- No large pivoting swept volume.
- More structure stayed nested at partial height than cascade.

Tradeoffs We Accepted

- Rope routing and tension needed careful setup.
- Guides had to stay aligned without excessive clamp load.
- Friction could prevent gravity-only return.
- Stage retention and end stops had to be checked during the build.

First Build Plan

- Build a three-stage CAD layout to test the maximum travel idea.
- Keep the stages captured with opposing lower bearing tabs.
- Check slide clearance before adding the full mechanism load.
- Power the elevator, record the first failure and change one cause at a time.

Robot Selection

----------Overview----------

Goal

To explain the importance for selecting the correct wheel size and justify our final decision to use 2.75-inch wheels for our first robot.

Selection

We have decided to choose the “Continuous Rigged Lift” design with 2.75-inch wheels operating at 450 rpm.

Justification

- The robot design chosen provides the strongest overall balance between scoring capability, reliability, speed, strength, stability and versatility.
- The 2.75-inch wheels were chosen because they provide fast acceleration, a lower centre of gravity, and greater precision, making them better suited to our focus on autonomous performance.
- We chose 450 RPM because it provides the best balance of acceleration, torque and control, making it effective for both offensive and defensive gameplay while maintaining precise autonomous movement.

Next Steps

- We will create an initial CAD model of the drivetrain to determine the layout of the gears, wheel positions, and cross-bracing.
- We will also create a CAD model of the arm and claw. However, we expect that the intake can be designed and built directly on the robot if necessary.
- We will document the entire CAD and construction process, ensuring that all documentation is relevant to the current game and robot design rather than previous seasons.

ELEVATOR BUILDING ELEMENTS

Six build cycles: the issue, what caused it, and what we needed to learn next.

CYCLE 1

THREE-STAGE CAD AND FIRST FRICTION

Cycle 1 - Issue

----------Issue 1 of 6----------

Goal

Explain the first motion problem before changing the elevator.

Initial Condition

- Three nested stages were assembled and wired.
- Opposing lower bearing tabs kept the stages captured.
- The tab positions preserved as much usable travel as possible.
- First powered movement checked whether the full mechanism could slide freely.

Reported Symptoms

- The elevator showed excessive sliding friction.
- Movement was difficult before the full mechanism load was added.
- Binding suggested the guides were being clamped against the C-channel.
- No time, current or trial count was recorded, so the result remained qualitative.

Why It Happened

Machined standoff length could vary slightly. If a standoff was short, or the fastener stack was over-tightened, the guide was compressed against the C-channel. This reduced the running clearance and increased friction across the nested stages. The next change therefore needed to isolate fastener tension as the cause before altering the full design.

Cycle 1 - What We Needed to Learn

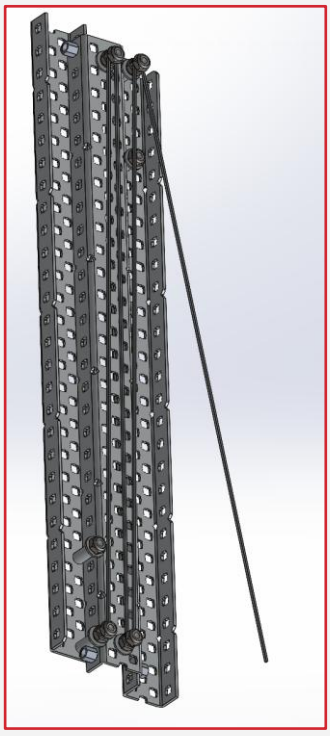
----------Learn 1 of 6----------

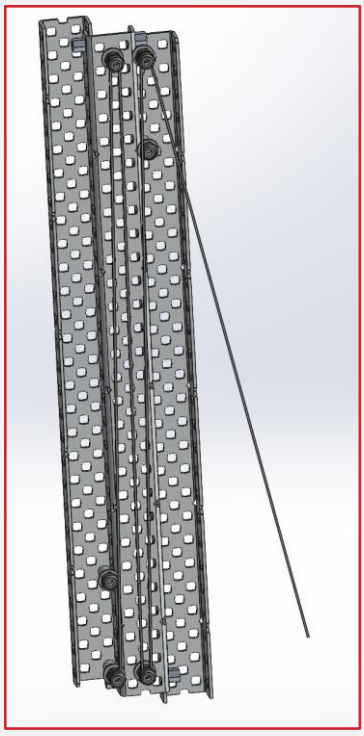
Goal

Record the question, change and learning that led into the next cycle.

What We Needed to Learn
QUESTION Was guide clamp load causing the friction? CHANGE Loosen the guide fasteners enough to restore clearance while keeping each guide secure. WHY THIS CHANGE It altered the suspected clamp load without rebuilding the elevator. LEARNING Motion improved, confirming fastener tension caused part of the drag. NEXT CYCLE Gravity return was still unreliable, so the lift needed positive retraction.

CYCLE PHOTO





CYCLE 2

CLEARANCE FIX AND POSITIVE RETURN

Cycle 2 - Issue

----------Issue 2 of 6----------

Goal

Explain why better guide clearance did not guarantee full retraction.

Initial Condition

- The three-stage elevator was retained for the next check.
- Guide fasteners were loosened to restore sliding clearance.
- The same powered extension and descent movement was repeated.
- The test focused on whether the lift could return fully after rising.

Reported Symptoms

- Extension improved after the clearance adjustment.
- Descent remained unreliable and the elevator could stop partly raised.
- Gravity did not provide a consistent pull-down force against the remaining resistance.
- The comparison was observational; no return time or success rate was recorded.

Why It Happened

Gravity still had to overcome guide friction, rope resistance and small alignment changes. When those combined forces were greater than the available downward force, the slide stopped before reaching the bottom. Better clearance solved only part of the problem, so the elevator needed an active return path rather than relying on its own weight.

Cycle 2 - What We Needed to Learn

----------Learn 2 of 6----------

Goal

Show how the remaining return problem defined the next build change.

What We Needed to Learn
QUESTION How could the elevator be pulled down positively instead of relying on gravity? CHANGE Keep the clearance adjustment and add two retraction strings tied to the high-strength shaft. WHY THIS CHANGE The strings could apply return force throughout the descent. LEARNING TARGET Determine whether active pull-down removed the incomplete-return problem. NEXT CYCLE Test the twin-string layout after chain and mechanism load were added.

CYCLE PHOTO



CYCLE 3

TWIN-STRING LOADED FAILURE

Cycle 3 - Issue

----------Issue 3 of 6----------

Goal

Explain why the first positive-return system failed when the mechanism was loaded.

Initial Condition

- Two pull-down strings were tied to the high-strength shaft.
- Rope anchors shared screws with bearing washers.
- Chain and mechanism load were added to the three-stage elevator.
- The loaded test checked whether positive return still worked in the complete mechanism.

Reported Symptoms

- The loaded elevator struggled to move through its travel.
- Bearing washers contacted the moving slide surfaces.
- The contact created severe friction during the test.
- Payload, motor current and trial count were not recorded, so only the failure mode is claimed.

Why It Happened

Rope tension acted through a fastener that also positioned a bearing washer and set guide clearance. The tension pulled the washer toward the moving slide until the surfaces contacted. Tightening the rigging could therefore worsen guide friction. The two functions needed separate hardware so adjustment of one could not disturb the other.

Cycle 3 - What We Needed to Learn

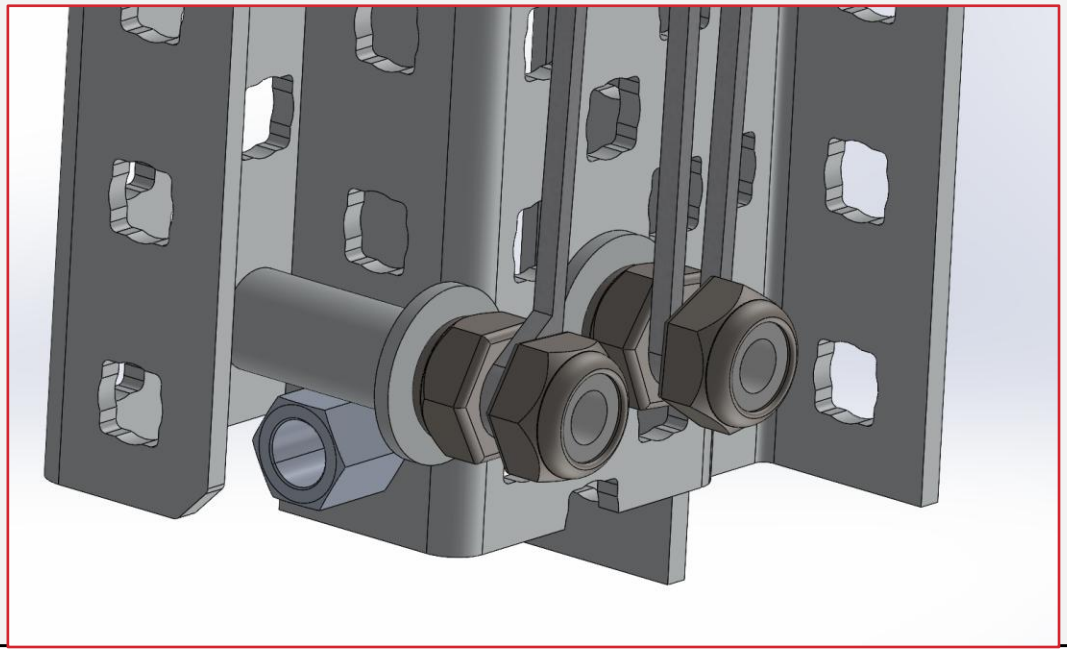
----------Learn 3 of 6----------

Goal

Record what the failure taught us before rebuilding the elevator.

What We Needed to Learn
QUESTION Would separating the anchors and reducing the number of stages remove the loaded failure? CHANGE Reject the shared-anchor arrangement and remove the third stage. WHY THIS CHANGE It separated the rigging load while reducing moving structure and sliding interfaces. LEARNING Rope loads must be independent from hardware that sets guide clearance. NEXT CYCLE Rebuild and evaluate the elevator as a two-stage system.

CYCLE PHOTO



CYCLE 4

TWO-STAGE REBUILD

Cycle 4 - Issue

----------Issue 4 of 6----------

Goal

Explain the tradeoff created by removing the third elevator stage.

Initial Condition

- The third elevator stage was removed from the mechanism.
- The same vertical guided-motion concept was retained.
- The simpler two-stage elevator was reassembled and operated.
- The comparison focused on usable motion rather than maximum theoretical reach.

Reported Symptoms

- The earlier three-stage version had struggled under mechanism load.
- The two-stage rebuild was reported as faster with more usable torque.
- Motor response was described as better during the qualitative comparison.
- The accepted drawback was reduced maximum travel, which was not measured.

Why It Happened

Removing one stage reduced moving structure, sliding interfaces and accumulated alignment demand. The motors had less mass and friction to overcome, improving the reported response. The tradeoff was a shorter maximum extension. The design decision therefore depended on whether reliable movement was more valuable than unverified extra reach.

Cycle 4 - What We Needed to Learn

----------Learn 4 of 6----------

Goal

Show why the shorter elevator became the retained direction.

What We Needed to Learn

QUESTION Was dependable motion more valuable than the extra travel of the third stage?
CHANGE Retain the two-stage structure and accept the reach tradeoff.
WHY THIS CHANGE Fewer moving interfaces reduced the sources of friction and misalignment.
LEARNING The team preferred the reported speed, usable torque and motor response of the simpler system.
NEXT CYCLE Improve the return path without reintroducing the failed shared anchors.

CYCLE PHOTO



CYCLE 5

CONTINUOUS RETRACTION ROPE

Cycle 5 - Issue

----------Issue 5 of 6----------

Goal

Explain why the failed twin strings were replaced by one continuous rope.

Initial Condition

- The two-stage elevator was retained as the base design.
- The separate tied pull-down strings were removed.
- One rope was passed through drilled spool holes as a linked path.
- The new route was kept away from the screws that controlled guide clearance.

Reported Symptoms

- The earlier twin-string route had loaded the bearing hardware.
- Separate tied anchors added more points where tension could become unequal.
- The return system needed to pull both directions consistently without disturbing guide alignment.
- Exact anchor clearance and repeated-cycle data were not recorded.

Why It Happened

The previous strings depended on separate anchors near hardware that also controlled the sliding guides. A single continuous route linked the retraction action and removed the failed washer-loading path. This reduced the number of independent tension points, while keeping positive pull-down rather than returning to unreliable gravity descent.

Cycle 5 - What We Needed to Learn

----------Learn 5 of 6----------

Goal	Record what the continuous path changed and what still needed checking.

What We Needed to Learn
QUESTION Would one linked rope give the two-stage elevator a more dependable positive return? CHANGE Drill the spool, pass one continuous rope through it and separate the route from guide fasteners. WHY THIS CHANGE One linked path coordinated retraction and removed the known washer-loading route. LEARNING Retraction was reported as better and faster than the two-string arrangement. NEXT CYCLE Package the retained rope path inside a compact low-profile spool.

CYCLE PHOTO





COMPETITION RECAP & ANALYSIS

EVALUATING PERFORMANCE AT SHORE AND IDENTIFYING IMPROVEMENTS

Competition Review/Analysis

----------Shore Competition----------

Goal	Reflect on our overall performance at the competition, so we can improve on our robot and strategies.

Alliance	83 – 8 Win	105 – 46 Win	32 – 56 Lose	60 – 26 Win	60 – 8 Win
Skills	Auto	1st – 8			
	Driver	1st – 28	2nd – 23	3rd - 18	

----------Main Rules----------

At the start of the competition, the Coaches, Judges, and attending ran through how the game is played, game rules, strategies and violations that can be given.

How the game is played + Strategies	Game Rules + Violations
The games is played by intaking, scoring and switching toggles. These all earn points that accumulate to a high score. Skills – Played as an individual team, with no opponent. Autonomous – A 15 second period, the robot aims to score and change toggle colours to gain alliance bonus. Alliance –Cooperate with your teammate to score high points.	The robot must be end within contracted form, within 18” x 18” x 18” Violations include: - Break of game elements - Holding more than one pin and one cup - Knocking over a stack of the opponent's alliance or neutral goal

Competiton Recaps

----------Overview----------

Goal

Review our performance at the Shores competition and identify our strengths, weaknesses and areas for improvement.

Overview

The Shores competition provided an opportunity for our team to test our robot in a competitive environment and evaluate how well our design, programming and driving strategy performed under match conditions. Throughout the competition, we were able to identify several strengths in our robot, while also discovering areas that required further development. Our team adapted throughout the competition. Our driver improved their driving technique over the course of the event, particularly when approaching game pieces, which reduced the number of pieces being accidentally moved or knocked away.

Challenges

- One of the main challenges throughout the competition was the reliability of our **toggle mechanism**. When robots made contact with the field perimeter, the field could be pushed out of position, which affected the position of the toggle. This created difficulties for both our match autonomous and Skills autonomous routines because our programming relied on the field elements being in their expected positions.

Achievements

- Overall, we performed strongly throughout the competition, finishing with a **5th-place ranking** and reaching the **semifinals**. We were also awarded the **Innovate Award**, recognising the design and development of our robot.

Skills Recaps

----------Overview----------

Goal

Evaluate our Skills performance at Shores to identify what went well, what limited our scores, and what improvements should be made before the upcoming competition.

Analysis

- Our Skills scores did not accurately represent the full potential of the robot because:
- The toggle mechanism created a significant limitation.
- It was not reliable when the field was displaced.
- This prevented us from consistently scoring all available points.

- The Driver Control attempts were particularly affected because:
- The robot was unable to reliably interact with the toggle, restricting the number of points that could be scored.
- The driver had to make adjustments when the field was out of position.

- Despite this, the driver was able to overcome some issues by knocking back pieces.

- Our best Driver Control score was **28 points**.

- The Autonomous Skills attempt was also affected because:
- Our autonomous routine depended on consistent field positioning.
- When the field was displaced, the robot's programmed movements were no longer perfectly aligned with the field elements.

- This reduced the accuracy of the autonomous routine.

- Our Autonomous Skills attempt scored **8 points**.

Skills Results

----------Overview----------

Goal

Evaluate our Skills performance at Shores to identify what went well, what limited our scores, and what improvements should be made before the upcoming competition.

Attempt	Type	Score	Notes
1	Driver Control	28	Toggle was working well when field was in place.
2	Driver Control	23	Toggle started struggling but still managed to work.
3	Driver Control	18	The field was moving, making it difficult to change toggle colours.
1	Autonomous	8	Field/toggle positioning affected autonomous performance
2	Autonomous	Nil	Not enough time for more attempts.
3	Autonomous	Nil	Not enough time for more attempts.

Overall
Our Skills performance was affected significantly by the reliability of the toggle mechanism. We completed **three Driver Control Skills attempts** and **one Autonomous Skills attempt**. We came 10th in skills.

Qualification Performance

----------Overview----------

Goal

Evaluate our qualification match performance at Shores to identify our strengths, areas for improvement, and how consistently our robot performed across different match situations.

Analysis

- We finished the qualification rankings in **5th place**, which was a strong result for our team.
- For much of the competition, we were positioned within the **top three**, showing that we were performing consistently against other teams.
- We eventually dropped to 5th later in the qualification rounds as other teams improved their scores.
- Finishing 5th demonstrated that our robot was competitive and capable of achieving strong results throughout the competition.
- Our strong early performances, including our **105-point match**, helped establish a high ranking.
- The 105-point match showed that our robot could score a large number of points when our strategy and robot performance worked effectively.
- We were able to maintain a competitive position despite some issues with the robot during the competition.
- Our team also adapted our driving and strategy between matches based on what we observed from opponents and alliance partners.
- Although we eventually dropped from the top three, finishing **5th** gave us a strong position heading into the elimination rounds.
- Overall, our qualification ranking showed that our robot design, driving and strategy were effective enough to compete at a high level.

Qualification Performance

----------Overview----------

Goal

Evaluate our qualification match performance at Shores to identify our strengths, areas for improvement, and how consistently our robot performed across different match situations.

Strengths

- **Strong Early Performance:**
- We maintained a high ranking for much of the competition.
- Our strong early matches helped us remain within the top three for a significant portion of the qualification rounds.

- **High Scoring Potential:**
- Our robot demonstrated that it could achieve scores above **100 points**.
- Our **105-point match** showed that the robot could perform effectively when our strategy and mechanisms worked together.

- **Competitive Consistency:**
- We finished with a positive **4–3 qualification record**.
- This showed that we were able to achieve successful results across multiple qualification matches.

- **Adaptability:**
- The team was able to adjust our driving and strategy throughout the competition.
- We responded to different opponents and alliance partners to improve our performance during matches.

Qualification Performance

----------Overview----------

Goal

Evaluate our qualification match performance at Shores to identify our strengths, areas for improvement, and how consistently our robot performed across different match situations.

Improvements

- **Consistency:**
- Our scores varied significantly between matches.
- More consistent robot performance would help us maintain a stronger ranking throughout the competition.

- **Mechanism Reliability:**
- The toggle mechanism remained a major source of lost opportunities.
- Improving its reliability would allow us to score more consistently.

- **Autonomous Reliability:**
- Improvements to our autonomous program would help establish more consistent scores.
- The program should be able to perform accurately even when small changes occur on the field.

- **Field Adaptability:**
- The robot needs to better handle situations where field elements are slightly displaced.
- Designing mechanisms and programming that can tolerate small changes in field position would improve reliability.

Match Results

----------Overview----------

Goal

Evaluate our qualification match performance at Shores to identify our strengths, areas for improvement, and how consistently our robot performed across different match situations.

Match	Our Score	Opponent Score	Result
1	83	8	Win
2	105	46	Win
3	32	56	Loss
4	60	26	Win
5	60	8	Win
6	32	33	Loss
7	64	72	Loss

Strengths

----------Overview----------

Goal

Evaluate our Skills performance at Shores to identify what went well, what limited our scores, and what improvements should be made before the upcoming competition.

Strengths
• The robot was able to compete effectively in qualification matches and consistently complete its main scoring tasks. • We achieved several strong match scores, showing that our robot could perform reliably under competition conditions. • Our robot remained competitive throughout the qualification rounds and was able to contribute effectively to our alliance. • We were able to adapt our driving and strategy during matches depending on the actions of our opponents and alliance partners. • Our team worked effectively across different competition roles, including driving, strategy, scouting and technical support. • We were able to reach the elimination rounds, demonstrating that our performance throughout qualification was strong enough to progress further in the competition. • We received the **Innovate Award**, recognising the design and development of our robot. • Our overall qualification ranking was **5th**, showing that we performed consistently across the qualification matches.

Problems

----------Overview----------

Goal

Evaluate our Skills performance at Shores to identify what went well, what limited our scores, and what improvements should be made before the upcoming competition.

Problems

- The **toggle mechanism was unreliable** when the field was moved.
- Robots pushing or ramming into the field could cause the field to shift.
- When the field moved, the toggle could become misaligned.
- This affected:
- Autonomous routines.
- Skills autonomous.
- Driver control.
- Our ability to consistently change toggles.
- In elimination rounds, the toggle problem became particularly significant.
- The opposing alliance was able to control/hog some toggles, making it more difficult for us to change them.

Game Piece Problems

- Some game pieces were knocked backwards, interrupting cycles. Driver practice helped reduce this. The follow-up review also identifies the need for the claw to grip over both the pin and cup. Analyze the geometry as well as the approach technique.

Qualification Performance

----------Overview----------

Goal

Evaluate our qualification match performance at Shores to identify our strengths, areas for improvement, and how consistently our robot performed across different match situations.

Overall

After competing at Shores, we were able to identify several areas where our robot could be improved. Although the robot performed well overall, finishing **5th in qualification rankings, reaching the semifinals, placing 10th in Skills and receiving the Innovate Award**, the competition exposed some limitations that were not as obvious during testing.

Issue

The toggle was the most significant recorded issue: field movement changed alignment and affected operation. Game pieces could also be knocked backwards. Driver practice helped, but the follow-up review identifies two mechanical priorities: reduce lift friction and improve the claw's grip over both the pin and cup.

Effects

This became particularly problematic because the toggle was involved in our **autonomous and Skills strategies**. When the field position changed, our programmed movements could become less accurate, which reduced the reliability of our autonomous routines.

Overview Of Performance

----------Overview----------

Goal

Evaluate our qualification match performance at Shores to identify our strengths, areas for improvement, and how consistently our robot performed across different match situations.

Analysis

- The results show that our robot could achieve **high-scoring performances**, particularly in Matches 1 and 2, where we scored **83 and 105 points** respectively.
- Our strongest performance was Match 2, where we achieved our highest score of **105 points**.
- This demonstrated that the robot had the capability to score a large number of points when the robot, drivers and strategy worked effectively together.
- The 105-point match also helped establish our strong position in the qualification rankings.
- The graph shows that our scores **varied considerably between matches**, rather than remaining at a consistent level.
- Some matches were affected by **field conditions**, particularly when the field was displaced during play.
- **Opposing robots** could also affect our scoring by interfering with our movements or preventing access to scoring areas.
- **Mechanism reliability**, particularly the toggle mechanism, also created situations where we lost potential scoring opportunities.
- This indicates that while the robot was capable of strong performances, we needed to improve its **consistency and reliability**.
- Overall, the results demonstrate that the robot had strong scoring potential, but reducing the impact of external factors would help us achieve more consistent scores.

Overview Of Performance

----------Overview----------

Goal

Evaluate our qualification match performance at Shores to identify our strengths, areas for improvement, and how consistently our robot performed across different match situations.

Analysis

- Our robot was capable of scoring **over 60 points consistently in several matches**, demonstrating reliable scoring potential when our strategy and mechanisms performed effectively.
- Our highest score was **105 points in Match 2**, which was our strongest overall qualification performance.
- Our lowest score was **32 points**, occurring in Matches 3 and 6.
- The large difference between our highest and lowest scores highlights the **variation in our performance** across different matches.
- Our losses were relatively close in Matches 6 and 7, showing that **small improvements in scoring or strategy could have affected the final result**.
- Match 3 was our largest scoring loss, with a final score of **32–56**.
- Factors such as **mechanism reliability, field conditions, opposing robots and driving decisions** could have contributed to differences between matches.
- Overall, our robot demonstrated **strong scoring potential**, but improving consistency and reliability would help us achieve stronger results across every match.

Shore - Improvement Priorities

---------- Competition Analysis ----------

Goal

Turn the first competition review into specific mechanism changes.

Elevator / Lift - Reduce Friction

- Our post-Shore review identifies lift friction as an improvement priority.
- The earlier metal-contact problem gives us a reason to investigate the guide interface, not just increase motor effort.
- Delrin contact spacers are the new direction.

Claw - Retain Pin and Cup

- The claw needs to grip over both the pin and cup.
- Driver practice helped with pieces being pushed backwards, but the mechanical review must also examine entry clearance, contact shape and secure retention through lifting.

Keep the Competition Context

- The existing analysis also records toggle misalignment when the field moved.
- Keep that build improvement separate.
- The next cycle targets these mechanism issues rather than repeating the whole-robot design selection.

DESIGN CYCLE 2

REDESIGNING THE ROBOT FOR IMPROVED PERFORMANCE

Overview

----------Overview----------

Goal

The goal of this design cycle is to review our performance at the Shores competition, identify the main problems with our robot, and develop practical improvements.

Reason For Second Design Cycle
• Shore exposed opportunities for targeted improvements rather than a new whole-robot design. • Elevator / lift: reduce unwanted friction while maintaining stable guidance and useful reach. • Claw: grip over the pin and cup, with reliable entry, retention and release. • Toggle: improve operation when field position changes; keep this as build work. • Chain bar: check the revised claw and lift together through their movement range. • The following pages turn those priorities into options, design reasoning and planned checks.

Overview

----------Overview----------

Goal

The goal of this design cycle is to review our performance at the Shores competition, identify the main problems with our robot, and develop practical improvements.

Main Areas For Improvement
Elevator / Lift **Reduce unwanted guide friction. Investigate the selected Delrin contact spacers, rail alignment and clearance. Confirm performance under load.** **Claw** **Analyze how the jaws grip over both the pin and cup. Compare approach tolerance, retention through motion and clean release.** **Integration** **Check chain-bar clearance and claw orientation as the elevator changes height.** **Competition Preparation** **Keep the toggle fix in Build. Leave time for repeatable checks before the next competition; do not assume a CAD model proves reliability.**

IDENTIFY

IDENTIFYING IMPROVEMENTS

Challenges

----------Identify----------

Goal

Identify problems experienced during the Shores competition and determine areas of the robot that could be improved.

Challenges	Analysis
Toggle Reliability	• During the competition, opposing robots frequently contacted the field perimeter, causing the field to shift out of its intended position. • This affected the alignment and operation of our toggle mechanism, making it more difficult for the robot to interact with the toggle reliably.
Driver Adaptation	• The game-piece issue was partially overcome through additional driver practice and refinement. • By improving driving accuracy and learning to approach game pieces more carefully, the frequency of accidentally moving pieces was reduced.

Challenges

----------Identify----------

Goal

Identify problems experienced during the Shores competition and determine areas of the robot that could be improved.

Challenges	Analysis
Field Conditions	• Our robot was designed and programmed around the expected field dimensions and positions. • When the field was displaced, some of our programmed movements and distances were no longer as accurate, highlighting the importance of designing mechanisms and autonomous routines that can tolerate small changes in field conditions
Autonomous Impact	• The movement of the field had a significant effect on our autonomous routines. • Because the robot relied on consistent field positioning, even relatively small changes could cause the robot to miss the intended toggle or become incorrectly positioned. • This affected both our match autonomous and Skills autonomous runs.

Claw - Problem to Requirement

---------- Identify ----------

Goal

Translate the Shore game-piece issue into a pin-and-cup grip requirement.

Reference to Competition Analysis

- The Shore Problems page describes pieces being knocked backwards.
- Our follow-up review makes the requirement more specific: the claw must grip over the pin and cup rather than depend on a perfectly aligned approach.

What Must Change

- Analyze the opening around both game pieces, the curved contact profile and the vertical spacing of the plates.
- The jaws must accept the pieces, retain them during motion and release without catching.

Evidence of Improvement

- Compare pickups, retention and release with the same pieces and approach offsets.
- Record misses, slips and catches for each configuration.
- These checks are planned; no success rate has yet been supplied.

Elevator - Problem to Requirement

---------- Identify ----------

Goal

Reduce unwanted lift resistance without sacrificing guidance and stability.

Reference to Competition Analysis

- The post-Shore improvement review identifies lift friction as a priority.
- The team also reports an earlier metal-on-metal contact problem.
- We need to address the guide contact while retaining a stable path for the moving channels.

Contact Change to Investigate

- The team chose Delrin spacers to create plastic-on-metal contact at the intended guide interface.
- That material change must work with correct alignment and running clearance; it cannot compensate for rails that are clamped or skewed.

What Counts as Better

- Compare guide drag and full lift motion under controlled conditions, then inspect side play and wear.
- A smoother unloaded slide alone is not enough: the assembly also needs consistent travel with its working load.

Time Constraints

----------Identify----------

Goal

To identify the factors that influence the robot's performance and must be considered throughout the design process.

Constraint	Justification
Limited Development Time	• We have **limited time** to improve the robot, build new mechanisms and complete testing before the competition. • We need to **prioritise the most important changes** and ensure there is enough time to test and fix any issues. • Careful time management will be important to make sure the robot is **reliable and competition-ready**.
Competition Schedule	• The **UNSW competition is one month after Shores**, with the **Knox competition only one week later**. • This gives us a limited amount of time to improve, build and test the robot. • We therefore need to develop a **reliable and competition-ready robot quickly**.

Time Constraints

----------Identify----------

Goal

To identify the factors that influence the robot's performance and must be considered throughout the design process.

Constraint	Justification
Limited Testing Time	• New mechanisms need enough time to be **tested and refined** before competition. • Designs should be **built and tested early** so any problems can be identified quickly. • If a mechanism does not perform as expected, we need enough time to **modify or replace it**. • This is important because we cannot afford to spend too long on a design that may not be reliable.
Build and Programming Time	• A major redesign would require significant time to **manufacture, assemble and integrate** new mechanisms. • It would also require us to **reprogram autonomous, driver control and skills autonomous**. • Leaving less time for **testing and troubleshooting** before competition. • We need to focus on **targeted improvements** rather than completely redesigning the robot.

Robot Constraints

----------Identify----------

Goal

To identify the factors that influence the robot's performance and must be considered throughout the design process.

Constraint	Justification
Existing Robot Structure	• New mechanisms must be compatible with the **current robot structure** and mounting points. • Changes should connect securely to the existing frame without requiring major structural modifications. • New components must also work alongside the robot's existing mechanisms and systems.
Limited Space	• The **lift, claw and other mechanisms** must fit within the available space on the robot. • Mechanisms must not interfere with each other during movement or operation. • Components also need to remain accessible for **maintenance, repairs and adjustments**.

Robot Constraints

----------Identify----------

Goal

To identify the factors that influence the robot's performance and must be considered throughout the design process.

Constraint	Justification
Weight	• Additional C-channels, motors and structural components will increase the robot's overall weight. • Extra weight can reduce the robot's speed, acceleration and manoeuvrability. • Heavier mechanisms may require more motor power to move efficiently. • Only necessary components should be added to avoid making the robot unnecessarily heavy.
Center Of Gravity	• Heavy components should be positioned as low as possible to improve stability. • Mechanisms that extend upwards or forwards should be kept lightweight where possible. • Changes should not significantly shift the robot's weight to one side or make it unstable during movement. • Stability must be maintained during driving, turning, pushing and scoring.

Robot Constraints

----------Identify----------

Goal

To identify the factors that influence the robot's performance and must be considered throughout the design process.

Constraint	Justification
Motor Capacity	• We have a limited number of motors, meaning new mechanisms cannot require excessive additional motors. • Existing motors should be prioritised for essential mechanisms such as the drivetrain, lift and claw. • Motor use must be planned carefully to ensure enough power remains for the robot's most important functions.
Pneumatic Capacity	• Heavy components should be positioned as low as possible to improve stability. • Mechanisms that extend upwards or forwards should be kept lightweight where possible. • Changes should not significantly shift the robot's weight to one side or make it unstable during movement. • Stability must be maintained during driving, turning, pushing and scoring.

Robot Constraints

----------Identify----------

Goal

To identify the factors that influence the robot's performance and must be considered throughout the design process.

Constraint	Justification
Limited Opportunity For Repairs	• During competition, there is a **short amount of time between matches** to identify and fix problems. • Meaning we need to quickly **diagnose issues and make reliable repairs** without affecting our preparation for the next match. • The robot should be designed for **easy access, maintenance and quick adjustments**.
No Major Rebuild Between Competitions	• Since Knox is only one week after UNSW, we need to avoid designs that would require **another major rebuild** immediately after UNSW. • Any major changes should either **be completed and tested** before UNSW or be simple enough to adjust between competitions. • This gives us **limited time** to make **major improvements** after UNSW while still preparing the robot for Knox.

Robot Constraints

----------Identify----------

Goal

To identify the factors that influence the robot's performance and must be considered throughout the design process.

Constraint	Justification
Testing Before Competition	• Every significant modification must be **tested thoroughly** before being used in competition. • Changes need to be tested for **reliability, consistency and performance** in different situations. • This ensures we do not **introduce new problems** or use an unproven mechanism during a competition.
Existing Competition Strategy	• Changes must **support** our current strategy rather than requiring the entire robot strategy to be redesigned. • New mechanisms should **improve our existing scoring**, movement or game-plan without creating unnecessary changes to how the robot is used. • This allows us to improve the robot while keeping our current strategy familiar and reliable.

Performance Constraints

----------Identify----------

Goal

To identify the factors that influence the robot's performance and must be considered throughout the design process.

Constraint	Justification
Reliability	• Improvements must not reduce the reliability of mechanisms that are already working effectively. • Existing mechanisms should only be changed if there is a clear benefit to the robot's performance. • New modifications must be tested to ensure they do not interfere with current mechanisms. • Changes should maintain or improve the robot's consistency during matches and Skills runs. • Reliable components should be protected from unnecessary changes that could introduce new problems.
Autonomous Compatibility	• Sufficient time must be allowed to test and fine-tune autonomous routines after making changes. • Autonomous routines should be consistently tested to ensure the modifications have not reduced their reliability.

Performance Constraints

----------Identify----------

Goal

To identify the factors that influence the robot's performance and must be considered throughout the design process.

Constraint	Justification
Driver Control	• Controls should be simple and easy to understand during fast-paced matches. • New mechanisms should not require complicated combinations of buttons or controls. • The driver should be able to operate multiple mechanisms smoothly without losing focus on driving. • Mechanisms should respond consistently and predictably to driver inputs. • Any new controls should be practised and tested before competition to ensure the driver can use them confidently.

Performance Constraints

----------Identify----------

Goal

To identify the factors that influence the robot's performance and must be considered throughout the design process.

Constraint	Justification
Field Conditions	• The robot needs to perform consistently even when the field or game elements are not perfectly positioned. • Mechanisms should tolerate small changes in the position or alignment of cups, pins, goals and toggles. • The robot should not rely on perfectly precise positioning to collect or score game elements. • Mechanisms should be able to handle minor field changes caused by other robots during a match. • Designs should reduce the chance of missed collections, failed scoring attempts or misalignment. • Testing should include different positions and conditions to ensure the robot remains reliable.

Performance Constraints

----------Identify----------

Goal

To identify the factors that influence the robot's performance and must be considered throughout the design process.

Constraint	Justification
Cycle Time	• Mechanisms should not significantly slow down collection, scoring or movement around the field. • Collection should be quick and allow game elements to be picked up with minimal delay. • Scoring mechanisms should operate smoothly without requiring excessive setup or adjustment. • The robot should maintain good speed and manoeuvrability while using its mechanisms. • Mechanisms should avoid adding unnecessary weight that could reduce acceleration or movement speed. • The driver should be able to switch between collecting, scoring and moving efficiently during a match.

Constraints

----------Identify----------

Goal

To identify the factors that influence the robot's performance and must be considered throughout the design process.

Constraint	Justification
Time	Our biggest constraint is the **limited amount of time available before the UNSW competition**. We need enough time to design, manufacture, assemble and test any new mechanisms.
Existing robot	The new mechanisms must be compatible with our current robot. We cannot design a mechanism that requires the entire robot to be rebuilt or significantly changes the structure of the drivetrain.
Weight	Adding additional C-channels, motors or other components will increase the robot's weight. We need to make sure that any increase in weight provides a worthwhile improvement in performance.
Build Time	Any major redesign would require significant time to manufacture, assemble and integrate into the existing robot.
Programming Time	Any major redesign would require significant time to reprogram autonomous period, code the robot and program skills autonomous.

Constraints

----------Identify----------

Goal

To identify the factors that influence the robot's performance and must be considered throughout the design process.

Constraint	Justification
Space	The lift and claw must fit within the existing robot while still allowing the drivetrain and other mechanisms to operate without interference.
Reliability	The new design must be reliable under competition conditions. We do not want to replace a mechanism that already works with one that has significantly more opportunities for failure.
Autonomous Compatibility	Any changes to the lift or claw need to work with our autonomous routines. If the mechanism changes significantly, we will also need time to modify and retest our programming.
Driver Control	The mechanism needs to be straightforward for the driver to operate. A mechanism that is mechanically effective but difficult to control could negatively affect match performance.
Competition Conditions	The robot needs to perform consistently even when the field is slightly displaced or game pieces are not perfectly positioned. Our experience at Shores showed that designing only for perfect conditions can reduce competition reliability.

Criteria

----------Select----------

Goal

To compare multiple robot concepts against a defined success criteria and select the design that best meets our strategic, mechanical and competition requirements.

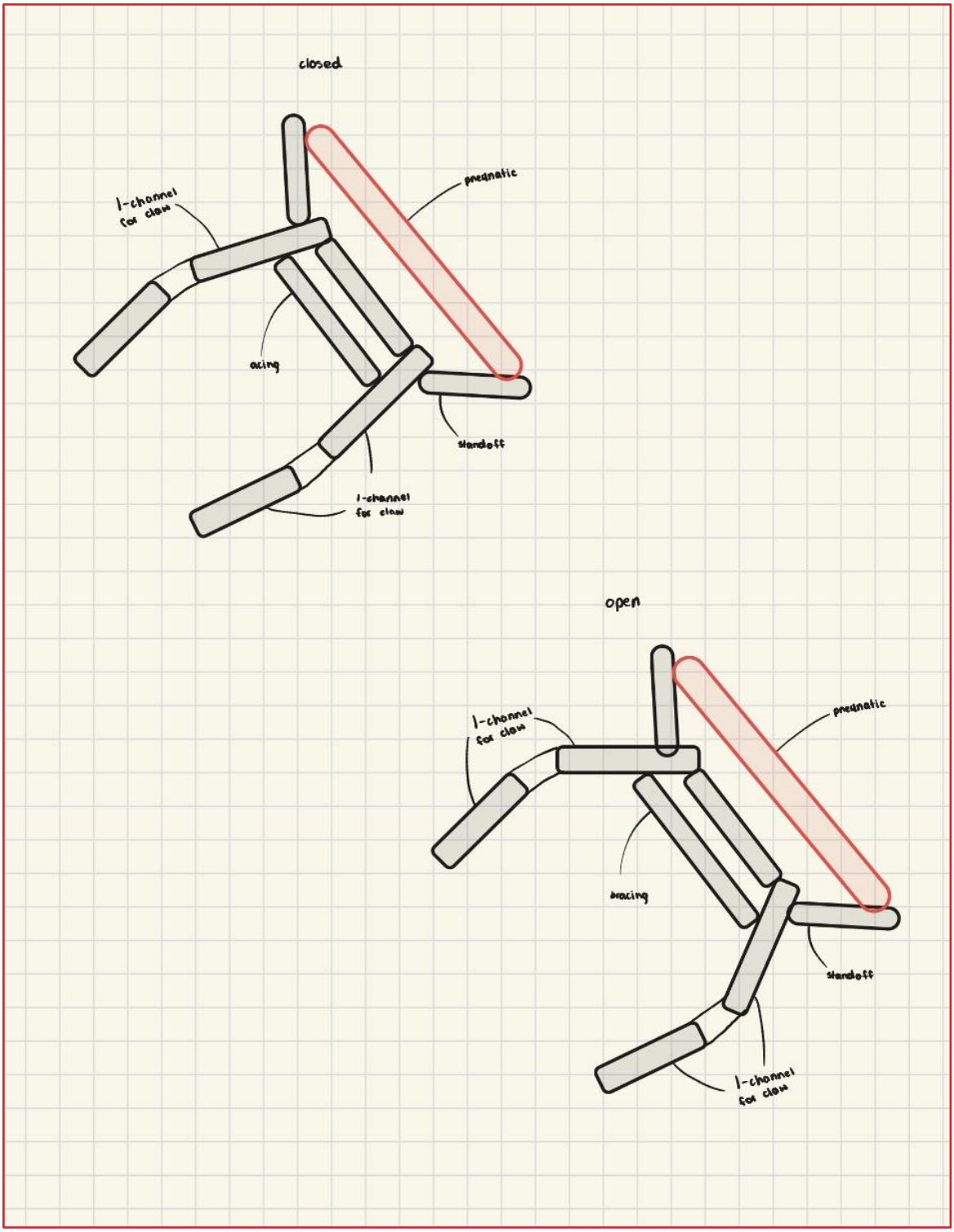
Criteria	What We Require
Scoring Capability	Ability to reach and score at **different heights** and **effectively interact** with important game pieces.
Reliability	Ability to **perform consistently** with **minimal risk of mechanical failure** during competition.
Speed And Efficiency	Ability to **complete scoring cycles quickly** while **minimizing unnecessary movement.**
Strength And Durability	Ability to **withstand collisions, defensive play and repeated operation** without bending or breaking.
Stability	Ability to maintain a **balanced center of gravity** and avoid tipping when mechanisms are extended.
Versatility	Ability to **perform multiple tasks** and **adapt** to different match situations.
Maintenance	The **ease of accessing, repairing ad replacing** components of the robot between matches.
Complexity	How difficult the robot design is to **build, program, operate and maintain.**
Compactness	How efficiently the mechanisms **use available space** without interfering with the drivetrain or other mechanisms.

BRAINSTORM

NEW ROBOT IDEAS FOR IMPROVED PERFORMANCE

Claw Design 1

----------Design Sketch----------



Claw Design 1

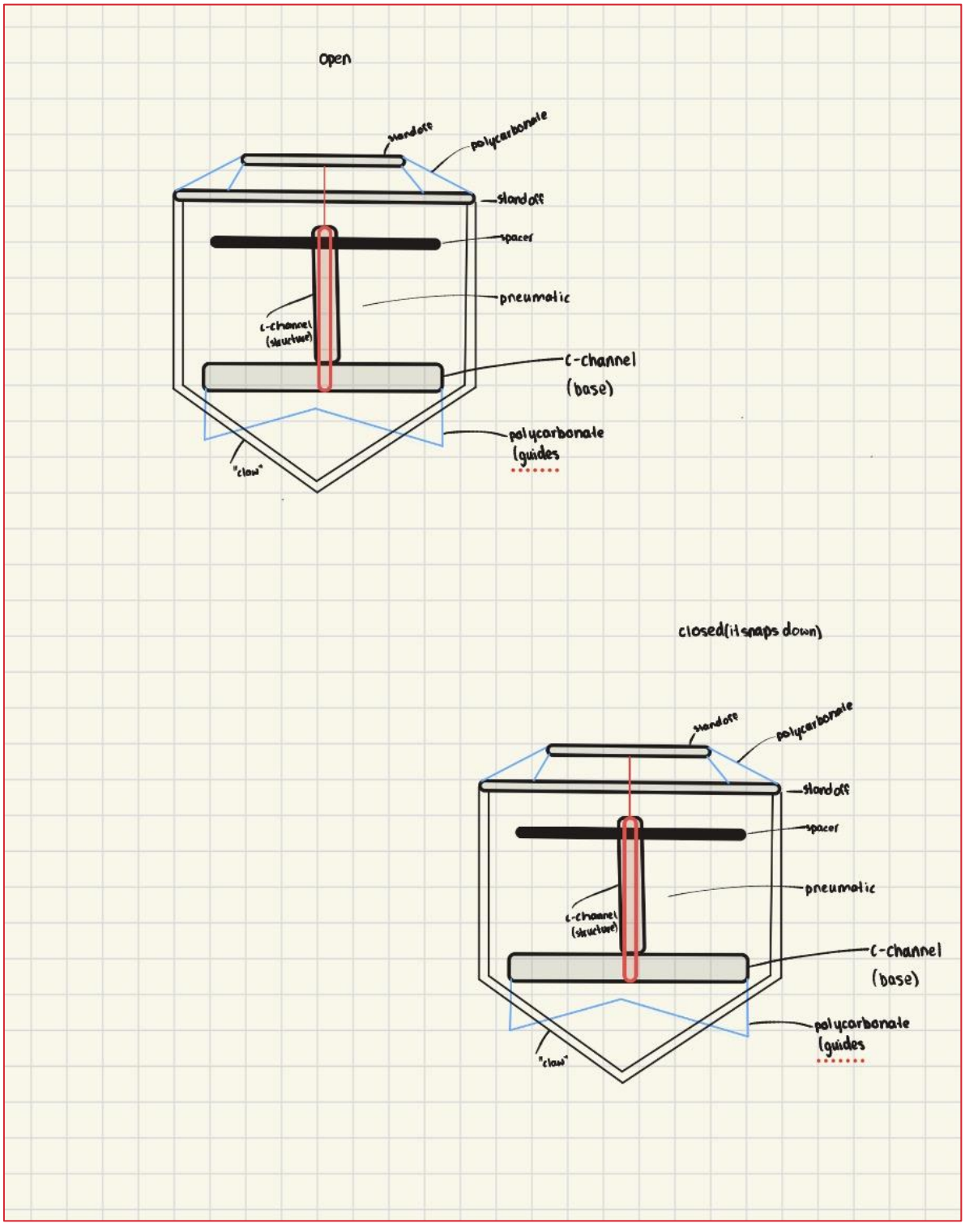
----------Analyze----------

Goal | Identify the strengths and weaknesses of a claw design

Strengths	Weaknesses
• **Large Intake Area:** The wide opening makes it easier to collect cups and other game objects without requiring extremely precise alignment. • **Efficient Object Collection:** The funnel shape helps guide game objects towards the centre of the mechanism, improving the consistency of collection. • systems. • **Strong Autonomous:** The large intake area provides a greater margin for error, which could improve the reliability of autonomous movements. • **Adaptable To Different Situations:** The ability to collect and reorient objects gives the robot greater flexibility when responding to different match and scoring situations.	• **Requires Careful Adjustment:** The mechanism will need to be tested and adjusted to ensure that the movement is smooth and consistent. functionality. • **Requires Sufficient Clearance:** The mechanism will need enough space to move through its full range of motion without interfering with other components. • **Additional Weight:** The structural components and flipping mechanism may add some weight to the robot, which will need to be considered when balancing the overall design. • **Pneumatic Dependence:** The flipping mechanism relies on the pneumatic system, meaning it will need to be designed to ensure consistent and reliable operation.

Claw Design 2

----------Design Sketch----------



Claw Design 2

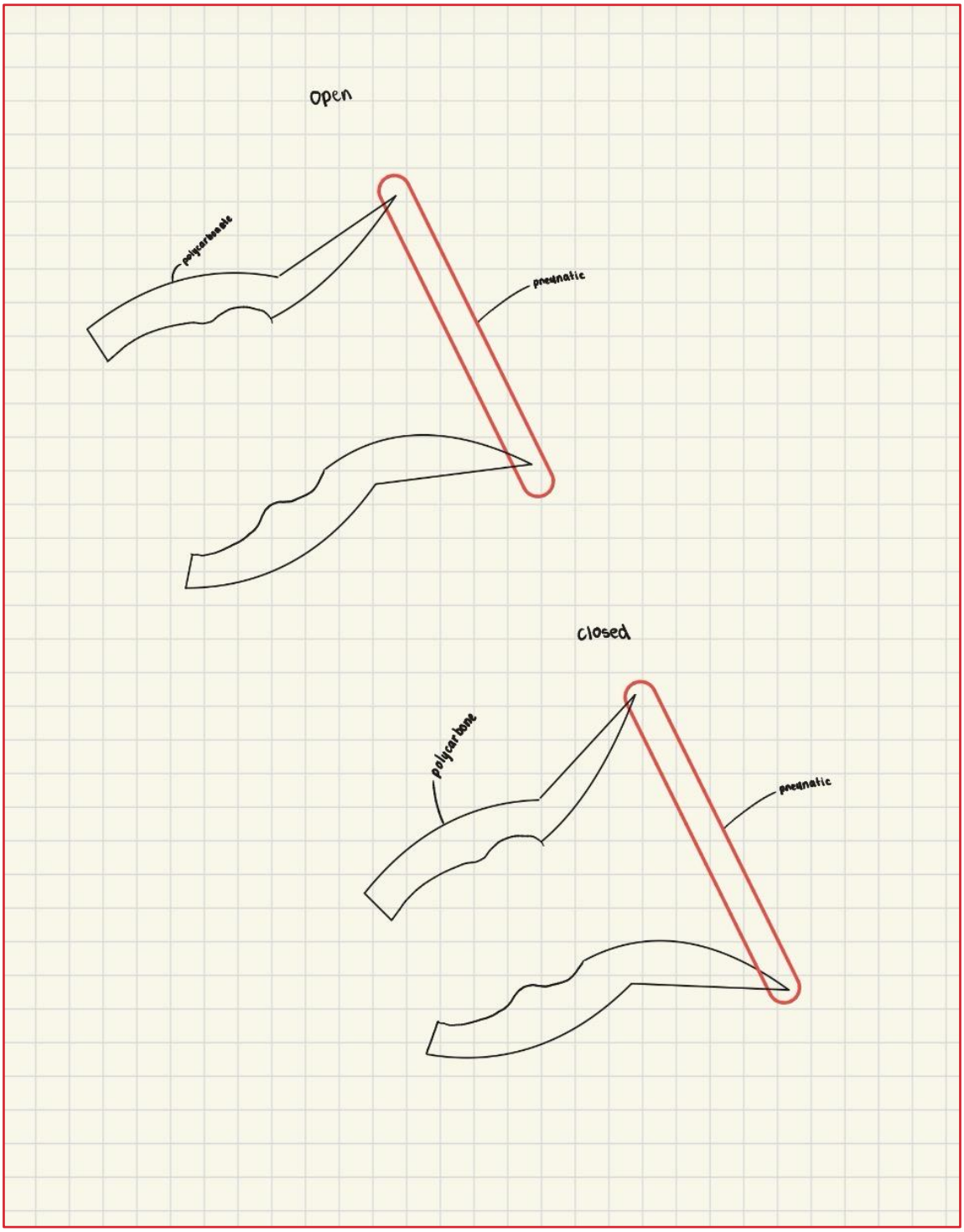
----------Analyze----------

Goal | Identify the strengths and weaknesses of a claw design

Strengths	Weaknesses
• **Collection Area:** The wide funnel opening provides a large target area, making it easier to collect game objects without requiring extremely precise alignment. • **Guides Objects Into Position:** The funnel shape naturally directs objects towards the centre of the mechanism, improving consistency of collection. • **Versatile Object Handling:** The motorised arms can provide additional control over objects, allowing the mechanism to assist with collecting, positioning, or manipulating pins and cups. • **Faster Collection:** Because the intake has a wide opening, the robot may be able to approach objects from a wider range of angles, reducing the time required for precise positioning.	• **Space Required:** The funnel and supporting structure take up a significant amount of space, which could limit room for other mechanisms. • **Additional Weight:** The large structural components, combined with the motorised arms, add weight to the robot. • **Complex Construction:** The mechanism requires multiple structural supports and carefully positioned components to ensure that the funnel, arms, and motors operate without interference. • **Risk Of Objects Becoming Stuck:** Depending on the exact dimensions and movement of the mechanism, cups or pins could potentially become jammed inside the funnel.

Claw Design 3

----------Design Sketch----------



Claw Design 3

----------Analyze----------

Goal | Identify the strengths and weaknesses of a claw design

Strengths	Weaknesses
•**Improved Game-Piece Control:** The claw is designed to fit around the game piece, providing more direct control. •**Polycarbonate Contact Surfaces:** Polycarbonate provides a lightweight and durable surface for gripping the object. •**Simple Mechanism:** The design avoids unnecessary mechanical complexity. •**Lightweight Construction:** Using polycarbonate can reduce the amount of heavy structural material required. •**Secure Grip:** The shaped contact points help hold the game piece securely. •**Good Driver Control:** The straightforward opening and closing movement should make the mechanism easy to operate. •**Suitable For Competition:** The mechanism provides a practical balance between functionality, simplicity and reliability.	•**Requires Testing:** The exact shape and positioning of the polycarbonate will need to be tested. •**Construction Precision:** The claw needs to be aligned correctly to ensure both sides make consistent contact. •**Limited Collection Area:** Unlike a large funnel intake, the claw does not provide a very large target area. •**Potential Wear:** The polycarbonate contact surfaces may need to be adjusted or replaced after repeated use. •**New Driver Familiarity:** The driver will need some practice to become comfortable using the new mechanism.

Lift Overview

----------Overview----------

Goal

Evaluate our qualification match performance at Shores to identify our strengths, areas for improvement, and how consistently our robot performed across different match situations.

Lift Design 1 - Existing Lift

The first option is to keep our current lift exactly as it is. This has the advantage of being the most familiar and already tested design. It would require very little additional construction or programming, which is particularly useful given the short timeframe before UNSW. However, keeping the lift unchanged would mean that we would not gain any additional height or capability from the next design cycle.

Lift Design 2 – Extended Lift

The second option is to extend our existing lift by adding an additional **C-channel section**. This would allow the lift to reach higher while maintaining the basic design that we already understand. It provides a compromise between improving the robot's capabilities and avoiding a completely new mechanism. The main consideration is that the additional C-channel will increase the height, weight and complexity of the lift. It will therefore require testing to ensure that the extension remains stable and moves consistently.

Lift Design 3 – New Design

The third option would involve replacing the current lift with a completely different lifting system. This could potentially provide greater functionality, but it would require considerably more design, construction and testing. Because UNSW and Knox are close together, this would create a significant development risk.

Claw - Improvement Options

---------- Brainstorm | Options for Review ----------

Goal

Compare three responses to the pin-and-cup retention requirement.

Option 1 - Refine the Existing Grip

- Adjust the existing opening and alignment to fit the actual pieces.
- This limits rebuilding, but may retain a narrow approach window.
- Compare how readily the pin and cup enter without being pushed away.

Option 2 - Shaped Upper / Lower Jaws

- The supplied CAD uses curved upper and lower plates joined by posts.
- Analyze this arrangement for capturing both shapes at the intended contact heights.
- Close fitting can improve location but makes clearance and release more sensitive.

Option 3 - Wider Entry Geometry

- Consider a wider lead-in feeding the gripping region.
- It may accept larger approach errors, but takes more space and can contact nearby pieces.
- This is a proposed alternative, not a claim that a new version was built.

Elevator - Improvement Options

---------- Brainstorm | Options for Review ----------

Goal

Compare ways to reduce lift resistance while preserving alignment.

Option 1 - Correct the Existing Setup

- Retain the contact parts and correct rail alignment, bolt loading and clearance.
- This is the smallest change, but does not directly replace the metal-contact interface identified in the review.

Option 2 - Delrin Contact Spacers

- Use the team-selected Delrin spacers at the intended contact points.
- The aim is lower unwanted sliding resistance while keeping the metal frame.
- Actual drag, side play and wear still require comparison.

Option 3 - Rolling Guide Redesign

- Consider a guide using rolling elements at selected contacts.
- This changes the contact arrangement but needs more packaging and alignment work.
- Keep it as an alternative for review rather than imply it was built or tested.

SELECT

SELECTING TARGETED MECHANISM IMPROVEMENTS

Criteria

----------Select----------

Goal

Use these criteria to assess targeted mechanism changes after Shore, rather than choose a replacement whole-robot concept.

Criteria	What We Require
Scoring Capability	Ability to reach and score at **different heights** and **effectively interact** with important game pieces.
Reliability	Ability to **perform consistently** with **minimal risk of mechanical failure** during competition.
Speed And Efficiency	Ability to **complete scoring cycles quickly** while **minimizing unnecessary movement.**
Strength And Durability	Ability to **withstand collisions, defensive play and repeated operation** without bending or breaking.
Stability	Ability to maintain a **balanced center of gravity** and avoid tipping when mechanisms are extended.
Versatility	Ability to **perform multiple tasks** and **adapt** to different match situations.
Maintenance	The **ease of accessing, repairing ad replacing** components of the robot between matches.
Complexity	How difficult the robot design is to **build, program, operate and maintain.**
Compactness	How efficiently the mechanisms **use available space** without interfering with the drivetrain or other mechanisms.

Mechanism Direction - Into CAD

---------- Select / Develop ----------

Goal

Carry the Shore priorities into mechanism-specific design analysis.

Elevator / Lift

- Develop the reported Delrin-spacer direction around the existing metal structure.
- Use the material research to explain the hypothesis, then examine alignment and clearance in CAD.
- Do not replace evidence with an assumed friction reduction.

Claw

- Develop the supplied shaped-jaw CAD against the explicit pin-and-cup requirement.
- Analyze the entry opening, contact heights and release path.
- The pictures establish the design direction, not validated grip performance.

Integration and Build

- Check the chain bar with the revised claw and lift through their shared movement envelope.
- Keep the toggle improvement in Build, with no toggle CAD stage.
- Advance using the stated criteria, not the removed whole-robot score table.

CAD DEVELOPMENT

Design Cycle 2 - turn the chosen concepts into detailed designs.

ELEVATOR / LIFT

Friction problem - contact theory - CAD - planned comparison

Elevator - Friction Hypothesis

---------- CAD | Theory and Rationale ----------

Goal

Explain why Delrin contact is worth investigating after Shore.

What the Research Supports [1]

- Delrin's manufacturer describes it as a low-friction engineering polymer and stresses that the mating surface and operating conditions affect performance.
- This supports considering it for a guide contact; it is not a result from our robot.

Apply the Idea to Our Lift

- Our hypothesis is that the new Delrin-on-metal interface will reduce unwanted guide resistance compared with our earlier metal-contact arrangement.
- Keep the metal channels for structure and use the spacer geometry to control the contact and clearance.

Limits of the Comparison [2]

- Plastic-on-metal is not automatically better in every situation.
- Grade, surface finish, load, speed and clearance matter.
- We must compare the actual assemblies before claiming a reduction; published steel test values are not our rail measurements.

[1] Delrin: Low Wear, Low Friction (manufacturer article)

[2] Delrin Design Guide, Module III, pp. 37-38

Elevator - Plan the Comparison

---------- CAD | Validation Plan ----------

Goal

Separate material effects from alignment, loading and drive effects.

Compare the Guide Interface

- On a supported, isolated guide, compare starting and running pull force before and after the contact change.
- Keep alignment, load, travel direction and speed consistent.
- Record the setup so bolt tension is not mistaken for a material benefit.

Check the Complete Lift

- Then compare complete lift travel using the same payload and control settings.
- Log extension/retraction time, sticking and position repeatability.
- Whole-lift results include drive and rigging effects, so they do not measure guide friction alone.

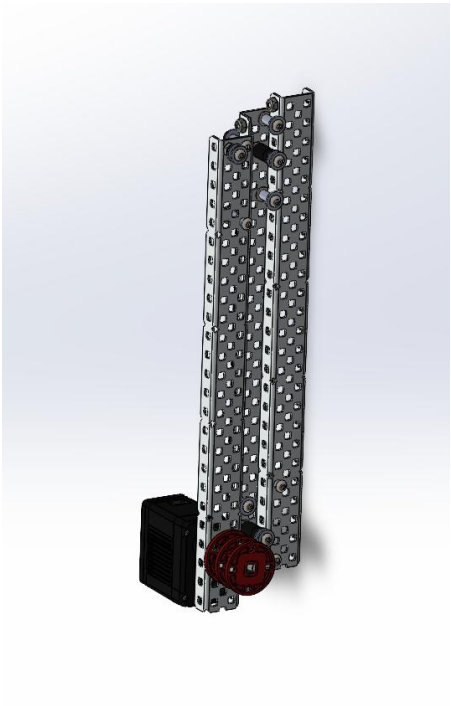
Use the Results to Revise

- Inspect spacer contact marks, wear and side play after repeated travel.
- Use those observations to revise spacing or alignment.
- This is a proposed comparison method; results are not recorded here.

Elevator - Guide Layout

---------- CAD DEVELOPMENT ----------

Goal Develop the nested-channel guide layout and its intended clearances in CAD.



Nested-channel CAD with guide hardware and base drive. Physical assembly evidence is in Build.

What the CAD Defines

The model places guide hardware along overlapping channels, with a motor and spool at the base. This separates the vertical guiding structure from the chain-bar mechanism carried above it.

Guide Interface in CAD

Plan the spacer contact and running clearance around the overlapping channels. Use the Delrin hypothesis to guide the design; the model alone does not establish smooth travel or load capacity.

Integration Checks

Rail alignment and clearance matter throughout the stroke. The screenshots show the intended layout, not measured travel or loaded deflection.

CLAW

Pin-and-cup requirement - grip analysis - CAD - planned checks

Claw - Pin and Cup Analysis

---------- CAD | Design Rationale ----------

Goal

Make the new claw answer the game-piece control issue from Shore.

Define the Capture Envelope

- Compare the open jaws with the real pin and cup, both separately and in the intended carried arrangement.
- Include an off-centre approach: the goal is to grip over the pieces without first pushing them away.

Analyze the Contact Geometry

- Use the curved jaw profiles and upper/lower plate spacing to examine where contact occurs on each shape.
- Check whether one piece can move even when the other is held.
- Do not assume one pictured CAD fit validates both.

Balance Entry Against Retention

- A larger opening can help entry, while closer surrounding geometry may help locate the pieces.
- The design needs both a usable approach window and secure retention, with enough clearance to open and release cleanly.

Claw - Retention and Release

---------- CAD | Validation Plan ----------

Goal

Plan a repeatable comparison of the old and revised grip.

Pickup Comparison

- Use the same pin, cup and approach offsets for each claw configuration.
- Record whether the jaws capture the pieces or push them away.
- Keep approach speed and actuator settings consistent; change one geometry feature at a time.

Retention Through Motion

- Move the held pieces through the intended lift and chain-bar positions.
- Observe slip, rotation and loss of either piece.
- Record failures separately from successful pickups so a good entry does not conceal poor retention.

Release and Revision

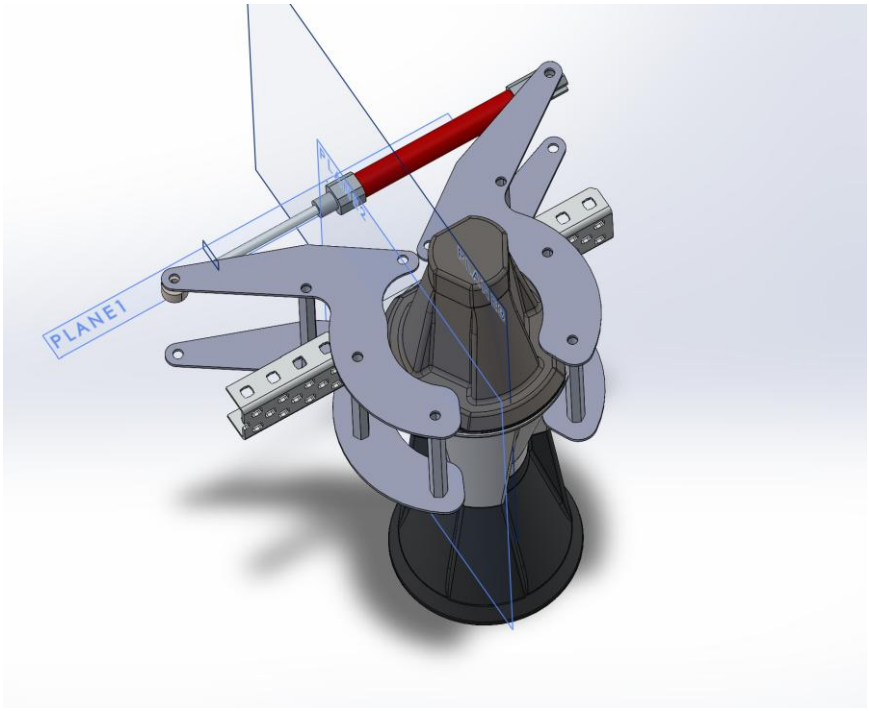
- Open the jaws at the intended placement position and check for catching, dragging or incomplete release.
- Use the failure location to revise contact shape or spacing.
- These are planned checks, not completed test results.

Claw - Capture Geometry

---------- CAD DEVELOPMENT ----------

Goal

Develop a claw that grips over both the pin and cup after the Shore review.



Curved jaw plates follow the modelled piece; posts connect the upper and lower plates.

Geometry in the Model

The CAD shows opposing curved jaws around the game piece. Each jaw has upper and lower plates joined by posts, with a channel mounting interface behind the grip.

Design Intent and Tradeoff

The surrounding profile is intended to locate the pin and cup. Analyze their separate contact heights and the intended carried arrangement. Close fitting may improve location but also reduces the allowance for off-centre entry.

Next Design Check

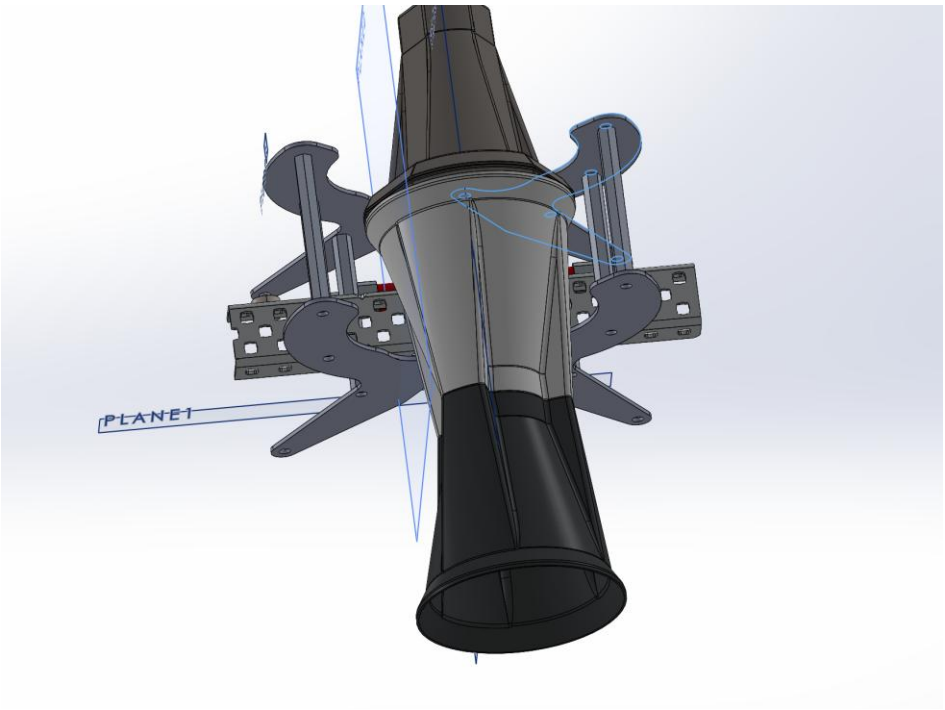
Entry, retention and release need to be checked with both pieces at different offsets. The CAD arrangement shows the intended geometry, not verified grip performance.

Claw - Actuation and Mounting

---------- CAD DEVELOPMENT ----------

Goal

Relate the jaw structure to its actuator and chain-bar connection.



The underside view exposes the paired plate levels, posts and rear mounting channel.

Actuator Layout

The upper view on the previous page shows a cylinder connected across the projecting jaw arms. This underside view shows how the upper and lower plates are tied together around the piece.

Assembly Interfaces

The rear channel is the structural connection to the arm assembly. The plate spacing and post positions must leave access for fasteners while keeping the gripping region clear.

Clearance Before Manufacture

The jaw sweep, cylinder ends and mounting bolts need clearance through opening and closing. A static screenshot cannot establish collision-free motion.

CHAIN BAR

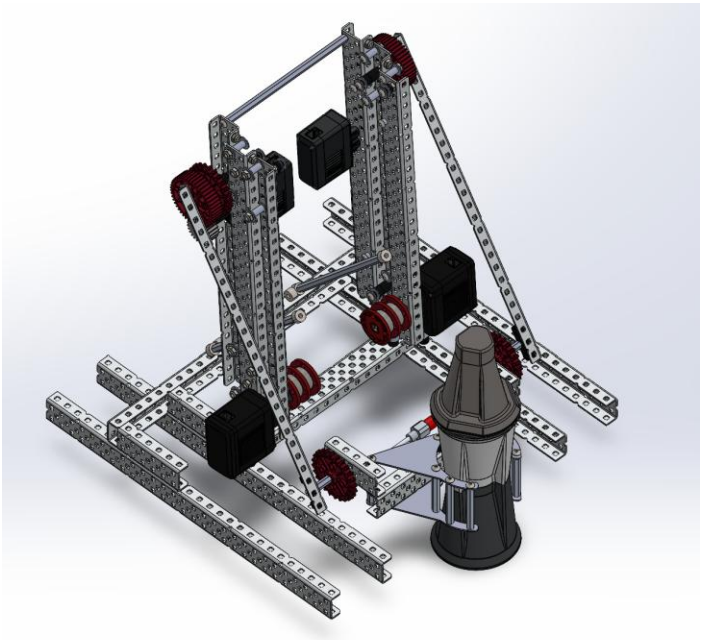
Integrate the revised claw and elevator through their movement range

Chain Bar - Robot Integration

---------- CAD DEVELOPMENT ----------

Goal

Show how the chain-bar assembly connects the elevator to the claw.



Assembly view shows paired arms, elevator-side pivots and the claw spanning the front ends.

Arrangement in CAD

Two long arms run from the elevator-side pivots to the claw mounting region. The CAD shows gears and drive components near the upper pivots, with the claw supported between the arm ends.

Interfaces to Resolve

The two sides need compatible pivot alignment and clearance around the elevator. The claw cylinder, mounting channel and fasteners must also remain clear as the arms change position.

Transmission Evidence Limit

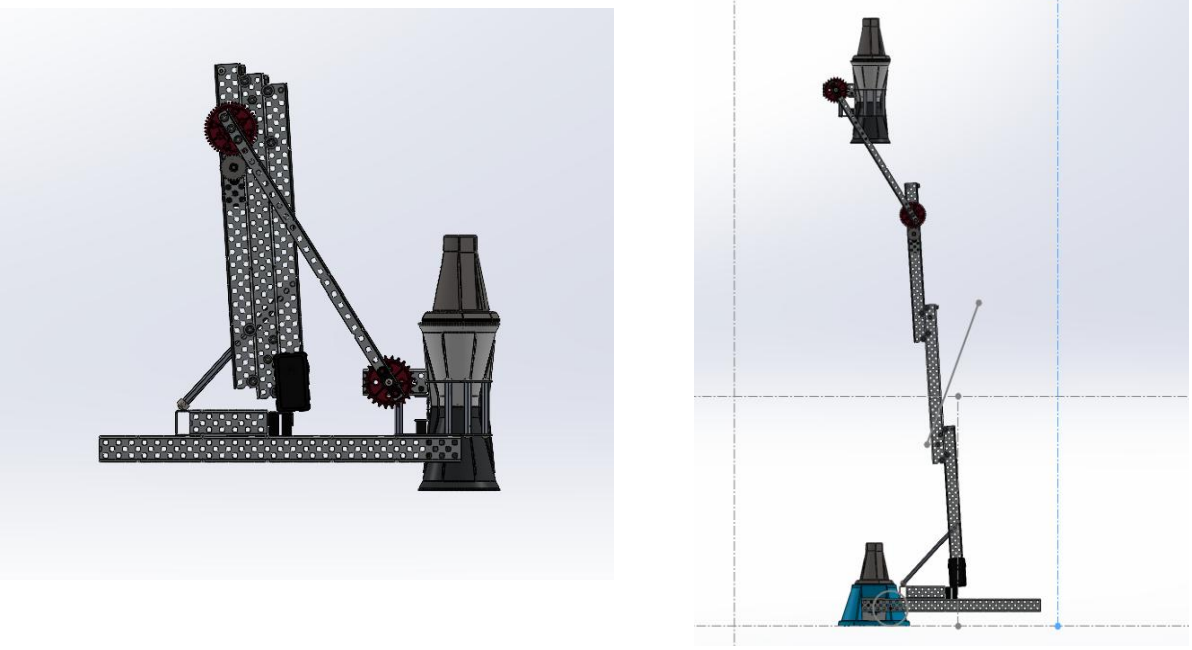
The supplied views do not show a complete chain path or specify a ratio. They show the assembly layout, not verified claw orientation through the full sweep.

Chain Bar - Position Study

---------- CAD DEVELOPMENT ----------

Goal

Use the supplied CAD poses to plan pickup and raised-position checks.



Left: low arm position. Right: raised elevator and arm pose. These are CAD states, not test results.

Why Both Views Matter

The low view places the claw near the base. The raised view combines elevator extension with arm rotation. Together they show why the elevator and chain bar must be considered as one moving assembly.

Checks Between the Poses

Check the intermediate sweep, not only the endpoints. Look for contact between the arms, claw, rails and game pieces; also leave room for the real transmission and any routed services.

Next Physical Comparison

The low and raised CAD poses show the intended movement range. They do not establish measured reach, repeatability or scoring success.

BUILD AND ASSEMBLY

Design Cycle 2 - manufacture, assemble and check the chosen designs.

Elevator - Guide Assembly

---------- BUILD AND ASSEMBLY ----------

Goal

Record the assembled elevator guide using the supplied team photo.



Assembled elevator guide supplied by the team. This is build evidence, not a CAD rendering.

Assembly Evidence
The photo shows the nested channels and spaced guide hardware assembled together. It supports the physical layout, but does not establish lifting capacity, smoothness or successful powered travel.

Contact Detail
The next Build page shows the guide-contact detail. The team identifies the revised spacers as Delrin; material identity comes from the team report, not appearance alone.

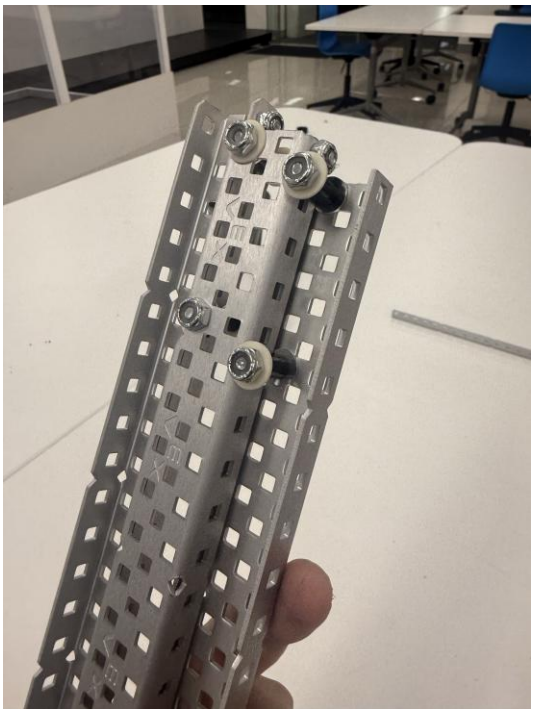
Checks Still Needed
The photo records the assembled guide layout. Rail alignment, clearance and powered travel are separate checks; no powered-test result is recorded on this page.

Elevator - Delrin Contact

---------- BUILD AND ASSEMBLY ----------

Goal

Replace the problematic metal-on-metal guide contact with plastic-on-metal contact.



Assembled guide detail. The team identifies the revised contact spacers as Delrin.

Problem and Change

The post-Shore review makes lift friction a priority. The earlier metal-contact problem led us to choose Delrin spacers for plastic-on-metal contact. The elevator CAD theory and comparison pages explain the intended benefit and how we will check it.

Reasoning Behind the Detail

The aim is to keep the metal structure guided without relying on direct metal rubbing at this interface. Spacer fit matters: too little clearance risks binding; too much allows unwanted movement.

What Still Needs Validation

The assembly photo shows the contact arrangement, not a measured reduction in friction or wear. Free and loaded travel would need comparison to establish the benefit.

COMPETITION RECAP & ANALYSIS

EVALUATING PERFORMANCE AT UNSW AND IDENTIFYING IMPROVEMENTS

Competition Review/Analysis

----------UNSW Competition----------

Goal	Reflect on our overall performance at the competition, so we can improve on our robot and strategies.

Alliance	0 – 85 Loss	42 – 13 Win	16 - 44 Loss	55 – 50 Loss
Skills	Auto			
	Driver	1st – 8	2nd – 18	3rd - 8

----------Main Rules----------

At the start of the competition, the Coaches, Judges, and attending ran through how the game is played, game rules, strategies and violations that can be given.

How the game is played + Strategies	Game Rules + Violations
The games is played by intaking, scoring and switching toggles. These all earn points that accumulate to a high score. Skills – Played as an individual team, with no opponent. Autonomous – A 15 second period, the robot aims to score and change toggle colours to gain alliance bonus. Alliance –Cooperate with your teammate to score high points.	The robot must be end within contracted form, within 18” x 18” x 18” Violations include: - Break of game elements - Holding more than one pin and one cup - Knocking over a stack of the opponent's alliance or neutral goal

Competiton Recaps

----------Overview----------

Goal

Review our performance at the UNSW competition and identify our strengths, weaknesses and areas for improvement.

Overview

The UNSW competition provided an opportunity for our team to test our robot in a competitive environment and evaluate how well our design, programming and driving strategy performed under match conditions. Throughout the competition, we were able to identify several strengths in our robot, while also discovering areas that required further development. Our team adapted throughout the competition. Our driver improved their driving technique over the course of the event, particularly when approaching game pieces, which reduced the number of pieces being accidentally moved or knocked away.

Challenges

One of the main challenges throughout the competition was the end effector flipping over to score. Previously, the gear ratio of the end effector struggled to lift it over the robot. To overcome this issue, we adapted by adding surgical tubing to the sides of the end effector, to create ease for the motors. However, the surgical tubing was not VEX approved meaning we couldn't bring it to comp, and we couldn't get VEX approved tubing before the competition, so we adapted by using rubber bands. This pulled the end effector too far down, creating a difficult circumstance for the robot to score. This created difficulties for both our match autonomous and Skills autonomous routines because our programming relied on the field elements being in their expected positions.

Achievements

- Overall, we performed strongly throughout the competition, finishing with a **8th-place ranking** and reaching the **semifinals**.

Skills Recaps

----------Overview----------

Goal

Evaluate our Skills performance at UNSW to identify what went well, what limited our scores, and what improvements should be made before the upcoming competition.

Analysis

- Our Skills scores did not accurately represent the full potential of the robot because:
- We didn't allow sufficient time before the competition to program both autonomous and driver control.

- The Driver Control attempts were affected because:
- The programming of driver control wasn't working despite working perfectly the night before.
- Despite this, the driver was able to have experience and improved throughout the day.

- Our best Driver Control score was **18 points**.

- The Autonomous Skills attempt was also affected because:
- We decided to prioritise the driver control programming to ensure that the correct height of cups could be reached during matches and we could still continue to compete.
- This meant that we didn't have time to program autonomous.

- Therefore, unfortunately we couldn't put down any Autonomous scores on the day.

Skills Results

----------Overview----------

Goal

Evaluate our Skills performance at UNSW to identify what went well, what limited our scores, and what improvements should be made before the upcoming competition.

Attempt	Type	Score	Notes
1	Driver Control	8	The first pin didn't go in so we couldn't continue the run.
2	Driver Control	18	The heights of the end effector weren't correct.
3	Driver Control	8	The first pin didn't go in so we couldn't continue the run.
1	Autonomous	Nil	Didn't have time to program autonomous
2	Autonomous	Nil	Didn't have time to program autonomous
3	Autonomous	Nil	Didn't have time to program autonomous

Overall

Our Skills performance was affected significantly by programming. We completed **three Driver Control Skills attempts** and **no Autonomous Skills attempt**. We came 17th in skills.

Qualification Performance

----------Overview----------

Goal

Evaluate our qualification match performance at UNSW to identify our strengths, areas for improvement, and how consistently our robot performed across different match situations.

Strengths

- **Improved Performance:**
- Our performance improved throughout the day as we became more comfortable with the robot, improved our driving and adapted to the competition environment.

- **High Scoring Potential:**
- Our robot demonstrated that it could achieve high scores
- Our **81-point match** showed that the robot could perform effectively when our strategy and mechanisms worked together.

- **Adaptability:**
- The team was able to adjust our driving and strategy throughout the competition.
- We responded to different opponents and alliance partners to improve our performance during matches.

Qualification Performance

----------Overview----------

Goal

Evaluate our qualification match performance at UNSW to identify our strengths, areas for improvement, and how consistently our robot performed across different match situations.

Analysis

- We finished the qualification rankings in **8th place**, which was a strong result for our team.
- For much of the competition, we were positioned within the **top five**, showing that we were performing consistently against other teams.
- We eventually dropped to 8th later in the qualification rounds as other teams improved their scores.
- Finishing 8th demonstrated that our robot was competitive and capable of achieving strong results throughout the competition.
- Our strong early performances, including our **81-point match**, helped establish a high ranking.
- This match showed that our robot could score a large number of points when our strategy and robot performance worked effectively.
- We were able to maintain a competitive position despite some issues with the robot during the competition.
- Our team also adapted our driving and strategy between matches based on what we observed from opponents and alliance partners.
- Although we eventually dropped from the top five, finishing **8th** gave us a good position heading into the elimination rounds.
- Overall, our qualification ranking showed that our robot design, driving and strategy were effective enough to compete at a high level.

Overview Of Performance

----------Overview----------

Goal

Evaluate our qualification match performance at UNSW to identify our strengths, areas for improvement, and how consistently our robot performed across different match situations.

Analysis

- Our robot was capable of scoring an average of 45 points, demonstrating reliable scoring potential when our strategy and mechanisms performed effectively.
- Our highest score was **81 points in Quarter Finals**, which was our strongest overall qualification performance.
- Our lowest score was **16 points**, occurring in Match 3.
- The large difference between our highest and lowest scores highlights the **variation in our performance** across different matches.
- Close losses, such as in Match 4(55 – 50) show that **small improvements in scoring or strategy could have affected the final result**.
- Match 6 was our largest scoring loss, with a final score of 86 - 33.
- Factors such as **mechanism reliability, field conditions, opposing robots, driving decisions, and programming** could have contributed to differences between matches.
- Overall, our robot demonstrated **strong scoring potential**, but improving consistency, reliability and programming would help us achieve stronger results across every match.

Match Results

----------Overview----------

Goal

Evaluate our qualification match performance at Shores to identify our strengths, areas for improvement, and how consistently our robot performed across different match situations.

Match	Our Score	Opponent Score	Result
1	0	85	Loss
2	13	42	Win
3	16	44	Loss
4	55	50	Loss
5	26	80	Win
6	86	33	Loss
QF	81	0	Win
SF	98	57	Loss

Strengths

----------Overview----------

Goal

Evaluate our Skills performance at UNSW to identify what went well, what limited our scores, and what improvements should be made before the upcoming competition.

Strengths
• The robot was able to compete effectively in qualification matches and consistently complete its main scoring tasks. • We achieved several strong match scores, showing that our robot could perform reliably under competition conditions. • Our robot remained competitive throughout the qualification rounds and was able to contribute effectively to our alliance. • We were able to adapt our driving and strategy during matches depending on the actions of our opponents and alliance partners. • Our team worked effectively across different competition roles, including driving, strategy, scouting and technical support. • We were able to reach the elimination rounds, demonstrating that our performance throughout qualification was strong enough to progress further in the competition. • Our overall qualification ranking was **8th**, showing that we performed consistently across the qualification matches.

Problems

----------Overview----------

Goal

Evaluate our Skills performance at UNSW to identify what went well, what limited our scores, and what improvements should be made before the upcoming competition.

Problems

- We hadn't given sufficient time to programming driver control and autonomous skills.
- This meant that our programming was not fully developed or tested before the competition.
- During the autonomous period, we were only able to change the toggle to our alliance colour.
- Although this didn't score any pins, we could still score 10 points from changing the toggle colours.
- However, we were unable to use autonomous to score pins or complete more advanced actions that could have increased our score.
- This also meant that we did not have a fully developed Autonomous Skills routine.
- As a result, we were unable to complete a proper Autonomous Skills run and missed out on opportunities to build our overall Skills score.
- Our Driver Skills programming also had issues and was not working as reliably as needed.
- This significantly impacted our overall Skills score because we could not consistently complete the planned actions within the time available.
- This showed us that programming needs to be given more time for development, testing and troubleshooting before future competitions.

COMPETITION RECAP & ANALYSIS

EVALUATING PERFORMANCE AT KNOX AND IDENTIFYING IMPROVEMENTS

Competition Review/Analysis

----------Knox Competition----------

Goal	Reflect on our overall performance at the competition, so we can improve on our robot and strategies.

Alliance	83 – 8 Win	98 – 48 Loss	31 – 41 Win	50 – 88 Loss	48 – 20 Win
Alliance	60 - 63 Win	110 - 93 Loss	85 - 15 Win	123 - 0 Win	120 – 45 Loss

----------Main Rules----------

At the start of the competition, the Coaches, Judges, and attending ran through how the game is played, game rules, strategies and violations that can be given.

How the game is played + Strategies	Game Rules + Violations
The games is played by intaking, scoring and switching toggles. These all earn points that accumulate to a high score. Skills – Played as an individual team, with no opponent. Autonomous – A 15 second period, the robot aims to score and change toggle colours to gain alliance bonus. Alliance –Cooperate with your teammate to score high points.	The robot must be end within contracted form, within 18” x 18” x 18” Violations include: - Break of game elements - Holding more than one pin and one cup - Knocking over a stack of the opponent's alliance or neutral goal

Competiton Recaps

----------Overview----------

Goal

Review our performance at the Knox competition and identify our strengths, weaknesses and areas for improvement.

Overview

The UNSW competition provided an opportunity for our team to test our robot in a competitive environment and evaluate how well our design, programming and driving strategy performed under match conditions. Throughout the competition, we were able to identify several strengths in our robot, while also discovering areas that required further development. Our team adapted throughout the competition. Our driver improved their driving technique over the course of the event, particularly when approaching game pieces, which reduced the number of pieces being accidentally moved or knocked away.

Challenges

- One of our main challenges was the chain bar not always having enough torque to consistently flip game pieces into the scoring position.
- The end effector also sometimes failed to hold the cup and pin in the exact position we intended.
- This could cause the game pieces to shift when being flipped, making scoring less consistent.
- We adjusted our driving technique to make the chain bar movements slower and more controlled.
- We also focused on positioning the game pieces correctly in the claw before flipping.
- These adaptations improved our consistency and allowed us to continue scoring effectively throughout the competition.

Achievements

- Overall, we performed strongly throughout the competition, finishing with a **8th-place ranking** and reaching the **semifinals**.

Skills Recaps

----------Driver Control----------

Goal

The goal of this page is to evaluate our Driver Control Skills performance at the competition, identify what contributed to our score, and determine what improvements could increase our performance in future competitions.

Preparation

- Before the competition, we spent time developing and practising our Skills strategy.
- We prioritised Driver Control because it was more reliable and achievable within the limited time available.
- Our main focus was:
- Improving driver consistency.
- Completing cycles efficiently.
- Controlling game pieces accurately.
- Making the most of the time available.
- We also used practice sessions to identify issues with the chain bar and end effector.

Performance

- Our Driver Control performance was one of the stronger parts of our Skills run.
- We achieved a **score of 48**, which we were relatively happy with.
- The driver was able to:
- Move efficiently around the field.
- Complete scoring cycles.
- Control the robot consistently.
- Adapt when mechanisms did not operate exactly as expected.
- Prioritising Driver Control was a good decision because our autonomous Skills program had become unreliable immediately before the competition.
- Instead of spending the limited Skills time repeatedly attempting an unreliable autonomous routine, we focused on achieving the highest score we could through Driver Control.

Skills Recaps

----------Driver Control----------

Goal

The goal of this page is to evaluate our Driver Control Skills performance at the competition, identify what contributed to our score, and determine what improvements could increase our performance in future competitions.

Issues

- One of the main problems during Driver Control was the position of the end effector on the chain bar.
- The end effector was sometimes **clamping too low**.
- This meant that the driver had difficulty getting the mechanism into the correct position to pick up and manipulate game pieces.
- Initially, we found that moving the arm up and down could help position the end effector correctly.
- However, this:
- Added extra movement.
- Took additional time.
- Made the scoring process less consistent.
- Reduced the efficiency of our cycles.
- On the day of competition, we found that even moving the chain bar up and down did not consistently solve the issue.
- This meant that the driver had to make additional adjustments during runs.

Strengths

- We were able to achieve a **48-point Driver Control score**.
- The driver remained relatively consistent despite the mechanical issues.
- We prioritised a reliable scoring method rather than taking unnecessary risks.
- The team adapted to problems during the competition.
- Our practice beforehand helped the driver understand how to operate the robot when the end effector was not perfectly positioned.

Skills Recaps

----------Autonomous----------

Goal

The goal of this page is to evaluate our Skills Autonomous performance, explain why our planned autonomous routine could not be used at the competition, and identify improvements needed for future competitions.

Preparation

- Before the competition, we had started developing a **Skills Autonomous program**.
- The program was progressing well during our initial testing.
- We were able to complete the planned movements and scoring sequence.
- Based on our testing, we estimated that the autonomous routine had the potential to score **68 points**.
- This would have been a very strong Skills Autonomous score.
- We continued developing the program with the aim of having it competition-ready.

Problems

- Unfortunately, **the day before the competition**, the autonomous program suddenly began malfunctioning.
- Movements that had previously worked correctly were no longer operating as expected.
- Different parts of the routine stopped functioning consistently.
- We were unable to determine exactly why the program had stopped working.
- At the same time, we were also experiencing problems with the **claw and end effector**.
- This made it difficult to determine whether problems were caused by:
- Programming.
- Mechanical issues.
- The end effector position.
- The interaction between the mechanisms and the programmed movements.

Skills Recaps

----------Autonomous----------

Goal

The goal of this page is to evaluate our Skills Autonomous performance, explain why our planned autonomous routine could not be used at the competition, and identify improvements needed for future competitions.

Decision

- By the time of the competition, we did not have enough time to completely troubleshoot and rebuild the autonomous routine.
- We therefore decided **not to use the full Skills Autonomous program**.
- Instead, we prioritised reliability and completed a much simpler routine.
- Our autonomous Skills attempt focused on **parking**, giving us **8 points**.
- Although this was significantly lower than the approximately 68 points our original routine had the potential to achieve, it was a more reliable option given the circumstances.

Reasoning

- We had limited time during the competition.
- We did not have enough attempts to repeatedly test an unreliable autonomous routine.
- Attempting the full routine could have resulted in:
- The robot failing to move correctly.
- Game pieces being missed.
- Time being wasted.
- A lower or unsuccessful Skills score.
- We therefore decided to prioritise our **Driver Control Skills performance**, where we had greater confidence in our ability to score.

Skills Recaps

----------Autonomous----------

Goal

The goal of this page is to evaluate our Skills Autonomous performance, explain why our planned autonomous routine could not be used at the competition, and identify improvements needed for future competitions.

Impacts

- Our final Autonomous Skills score was 8 points.
- This was considerably lower than the approximately 68 points we believed our program could achieve during testing.
- This had a significant effect on our overall Skills ranking.
- It showed how important it is to have a reliable autonomous program completed and thoroughly tested well before competition day.

Strengths

- The autonomous routine had shown strong potential during development.
- We had successfully tested parts of the routine before the final day.
- We recognised that the program was unreliable before using it in competition.
- We made a strategic decision to prioritise a reliable result rather than risk an unsuccessful attempt.
- We were still able to gain 8 points from the simplified parking routine.

Skills Recaps

----------Autonomous----------

Goal

The goal of this page is to evaluate our Skills Autonomous performance, explain why our planned autonomous routine could not be used at the competition, and identify improvements needed for future competitions.

Improvements

- Complete the autonomous program earlier.
- Allow more time between the final programming changes and competition.
- Test the entire routine repeatedly rather than relying on individual sections working correctly.
- Keep backup versions of working code.
- Identify whether mechanical or programming changes are causing failures.
- Test the autonomous routine after any major mechanical changes.
- Allow enough time for troubleshooting before competition day.

Summary

Our Skills Autonomous program showed strong potential during development and was estimated to be capable of scoring approximately 68 points. However, the program began malfunctioning the day before the competition, at the same time that we were experiencing issues with the claw and end effector. With limited time and Skills attempts available, we decided to abandon the full autonomous routine and prioritise Driver Control instead. We completed a simplified parking routine for 8 points, but this significantly reduced our overall Skills potential.

Skills Results

----------Overview----------

Goal

Evaluate our Skills performance at Knox to identify what went well, what limited our scores, and what improvements should be made before the upcoming competition.

Attempt	Type	Score	Notes
1	Driver Control	48	The first pin didn't go in so we couldn't continue the run.
2	Driver Control	38	The heights of the end effector weren't correct.
1	Autonomous	8	Didn't have time to program autonomous

Overall

Overall, our Skills performance was a positive experience, particularly in Driver Control. We completed two Driver Control attempts, scoring **41 points** in the first and **38 points** in the second, which showed that our driver could operate the robot consistently and effectively under competition conditions. We were very happy with these scores, especially considering the issues we experienced with the end effector. Our Autonomous Skills attempt was less successful, with a score of **8 points**, as our planned autonomous routine had stopped working reliably the day before the competition. Although this was lower than the approximately **68 points** our original program had the potential to achieve, we made the decision to prioritise a reliable parking routine rather than risk an unsuccessful attempt. Overall, our Driver Control performance was a strong part of the competition, while our main area for improvement is completing and thoroughly testing our autonomous program earlier before future competitions.

Qualification Offensive Performance

----------Overview----------

Goal

The goal of this page is to evaluate how effectively we used an offensive strategy to score game pieces and contribute points to our alliance.

Offensive Strategy

- Prioritised collecting and scoring game pieces early in the match.
- Used the robot's scoring mechanisms to complete cycles efficiently.
- Focused on scoring before moving into a defensive role.
- Once available game pieces had been scored, we could transition towards defending the Toggles.
- This allowed us to contribute points while also limiting opportunities for the opposing alliance.

Strengths

- We were able to score high numbers of points in several matches.
- Our highest score of **123 points** demonstrated our offensive potential.
- Match 1, Match 8 and QM showed that we could dominate the scoring when the alliance worked effectively together.
- We were able to adapt our role depending on what our alliance needed.

Improvements

- We could improve the consistency of our scoring cycles.
- Some matches had significantly lower scores, showing that our offensive performance was not always consistent.
- We needed to reduce the amount of time spent correcting positioning or dealing with unreliable field elements.
- Better communication with our alliance could help determine when to continue scoring and when to transition into defence.

Qualification Defensive Performance

----------Overview----------

Goal

The goal of this page is to evaluate how effectively we transitioned from an offensive role to defence and used our robot to limit the opposing alliance's scoring opportunities.

Defensive Strategy

- Score game pieces first when possible.
- Move towards important Toggles after completing scoring cycles.
- Prevent opponents from easily accessing or controlling Toggles.
- Use the robot's positioning to make it harder for opponents to score.
- Adapt our level of defence depending on the current score and alliance strategy.

Strengths

•We were able to change between offence and defence during matches.
•This made our robot more versatile.
•Defending Toggles gave us another way to contribute when there were fewer game pieces available.
•Our strategy allowed us to support different alliance partners with different strengths.

Improvements

- We could improve our decision-making about exactly when to switch from offence to defence.
- Better communication with the drive coach and alliance partner could make transitions faster.
- We could practise defensive positioning more to avoid wasting time moving between areas of the field.

Match Results

----------Overview----------

Goal

Evaluate our qualification match performance at Knox to identify our strengths, areas for improvement, and how consistently our robot performed across different match situations.

Match	Our Score	Opponent Score	Result
1	83	8	Win
2	98	48	Win
3	31	41	Loss
4	50	80	Loss
5	48	20	Win
6	60	63	Loss
7	110	93	Win
8	85	15	Win
QF	81	0	Win
SF	98	57	Loss

Match Statistics

----------Analysis----------

Goal

The goal of this page is to analyse the results of each qualification match and identify patterns in our scoring and overall consistency.

Statistics

- **5 wins and 3 losses**
- **Highest score:** 110 points
- **Lowest score:** 31 points
- **Average score: 70.6 points**
- **Average opponent score: 46 points**
- **Total points scored:** 565
- **Total points conceded:** 368
- We scored **more than our opponents in 5 of 8 matches**.
- Our highest-scoring match was **Match 7**, with 110 points.
- Our biggest winning margin was **Match 8**, where we won 85–15.

Overall Performance

- Our results showed that we were capable of scoring consistently at a competitive level.
- The variation between our highest and lowest scores shows that our performance could be affected by the alliance, opposing robots and field conditions.
- Despite some losses, we were able to recover and perform strongly in later matches.
- Our **5–3 record** resulted in a strong qualification performance.

Match Overview

----------Conclusion----------

Goal

The goal of this page is to analyse the results of each qualification match and identify patterns in our scoring and overall consistency.

Statistics

Overall, we performed well throughout the qualification matches, finishing with **5 wins and 3 losses** and an average score of **70.6 points per match**. Our strongest characteristic was our ability to play primarily offensively while still adapting to a defensive role when necessary. We usually focused on scoring game pieces first, then moved towards guarding Toggles to restrict the opposing alliance. Our highest score of **110 points** showed our strong scoring potential, while the lower-scoring matches demonstrated that we still needed to improve consistency. The close 60–63 loss also showed how important small scoring opportunities can be. Overall, our qualification performance demonstrated that our robot was competitive and versatile, but improving consistency, communication and the transition between offence and defence could help us achieve stronger results in future competitions.

PROGRAMMING

How the code coordinates homing, scoring and mechanism movement.

How the Code Fits Together

---------- PROGRAMMING ----------

One request becomes coordinated robot movement.

01 Choose the action

Driver controls or an autonomous routine request a pin, scoring pose or intake return.

02 Coordinate the mechanisms

ScoringSystem sets the lift, chain-bar and claw targets in the required order.

03 Read, update, drive

A separate control task updates the mechanisms, writes outputs and reports telemetry every 10 ms.

The same mechanism logic is shared by driver control and autonomous.

Establishing the Lift Zero

---------- PROGRAMMING ----------

The lift needs a known starting point before its height readings are useful.

01 Place the lift at the bottom

The current configuration has no enabled limit switches. Bottom placement is done by hand before the initial zero.

02 Set the encoder reference

zero_at_start() asks the control task to zero an unhomed lift. Once zeroed, another mode entry keeps that reference.

03 Move using that reference

Driver entry requests the intake pose after homing is established. A normal move to pin 0 uses the saved zero; it does not search for a switch.

Re-zeroing and returning to a zero-position target are different actions.

Why the Chain Bar Waits

---------- PROGRAMMING ----------

For normal upward moves to pins 4 and 5, the lift must arrive before the bar crosses upright.

01 Start the lift movement

The destination pin supplies the lift height and final chain-bar angle. Both controllers receive their targets.

02 Pause at the midpoint

The bar can approach upright, but the gate prevents it from crossing the 150-degree midpoint while the lift is travelling.

03 Release after lift arrival

Once the lift is within its target tolerance, the bar can continue to its requested angle. The release stays latched for that destination.

This rule is not applied to every pin. Descending moves, manual/coast modes and direct commands have separate handling.

Lowering Without Conflicting Moves

---------- PROGRAMMING ----------

A downward move only needs midpoint clearance when the bar must cross upright.

01 Check the direction

If the bar and its requested target are on opposite sides of upright, the lift holds its current height.

02 Bring the bar to upright

The descent gate releases when the bar is within its midpoint tolerance, or has crossed it between readings.

03 Continue the descent

Reaching the midpoint clearance band lets the lift lower. While descent clearance is active, the upward arrival gate is disabled so the gates do not wait on each other.

Same-side returns do not need an upright detour. Direct home and forced-pin moves deliberately bypass these gates.

The Main Movement Parameters

---------- PROGRAMMING ----------

These are source-code defaults, not measured performance results.

01 Midpoint: 150 degrees

This is the configured upright angle. The general chain-bar arrival tolerance is 7 degrees.

02 Lift arrival: 20 encoder ticks

The coordinator uses the lift target check before releasing a gated upward move. Tick values refer to motor encoders.

03 Bar home: 8 degrees

An explicit zero-angle command is mapped to an 8-degree offset. Home uses its own tighter 2-degree arrival tolerance.

Tunable settings may differ at runtime. The current build has limit switches disabled; configured travel bounds are not a verified physical envelope.

Choosing a Scoring Position

---------- PROGRAMMING ----------

The code chooses a mechanism pose from the scoring context.

01 Choose side, goal and pin

Front/back scoring, Alliance/Neutral goal type and pin number select the relevant configuration.

02 Look up the distance row

Distance tables pair a goal distance with lift ticks and a chain-bar angle. Between rows the lookup interpolates; outside the range it clamps.

03 Send the two targets

The coordinator requests the lift height and bar angle, then applies the movement sequencing rules.

A lookup table is a configuration, not proof of scoring accuracy. Some source rows are examples rather than measured calibration.

Scoring and Returning to Intake

---------- PROGRAMMING ----------

The scoring macro breaks a cycle into simple driver-controlled stages.

01 Prepare the scoring pose

Front scoring captures the selected pin. Back scoring prepares and holds one scoring pose.

02 Advance and release

Front and back scoring release on their next macro press. The coordinator holds the scoring pose during release.

03 Cancel or begin the next cycle

With a macro active, L1 cancels it, opens the claw and requests a direct return to intake. With no macro active, L1 only toggles the claw.

Autonomous helpers use the same mechanism coordinator. This describes the code sequence, not a recorded success rate.